

DATA MINING **VISUALIZATION**

(Many slides from Jeff Heer and Sam Lau)

What is visualization?

“Transformation of the **symbolic into the geometric**”

[McCormick et al. 1987]

“... finding the **artificial memory** that best supports our natural means of perception.” [Bertin 1967]

“The use of computer-generated, interactive, visual representations of data to **amplify cognition**.”

[Card, Mackinlay, & Shneiderman 1999]

Why Create Visualizations?

1. **Record** information

Blueprints, photographs, seismographs, ...

2. **Analyze** data to support reasoning

Develop and assess hypotheses

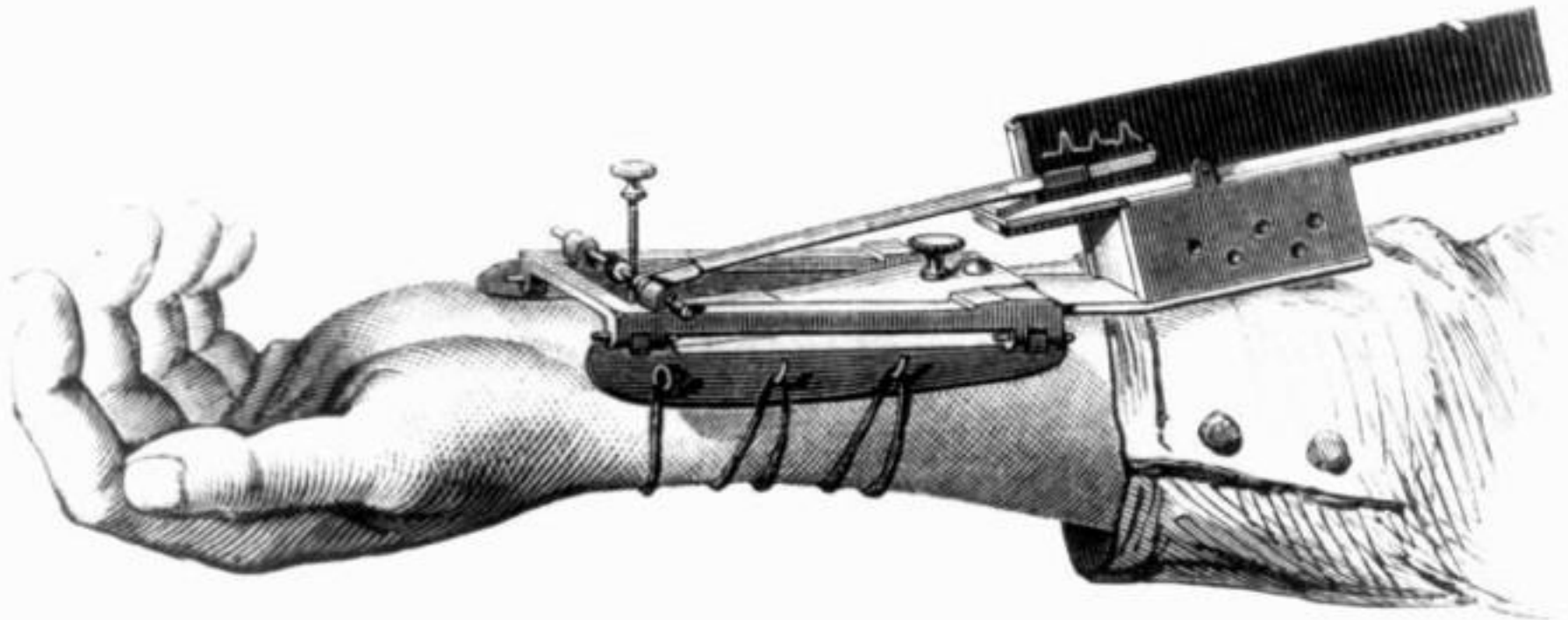
Discover errors in data

Find patterns

3. **Communicate** information to others

Share and persuade

Collaborate and revise



1.

Marey's **sphygmograph** in use,
1860. *La méthode graphique dans
les sciences expérimentales et
principalement en physiologie et en
médecine.*

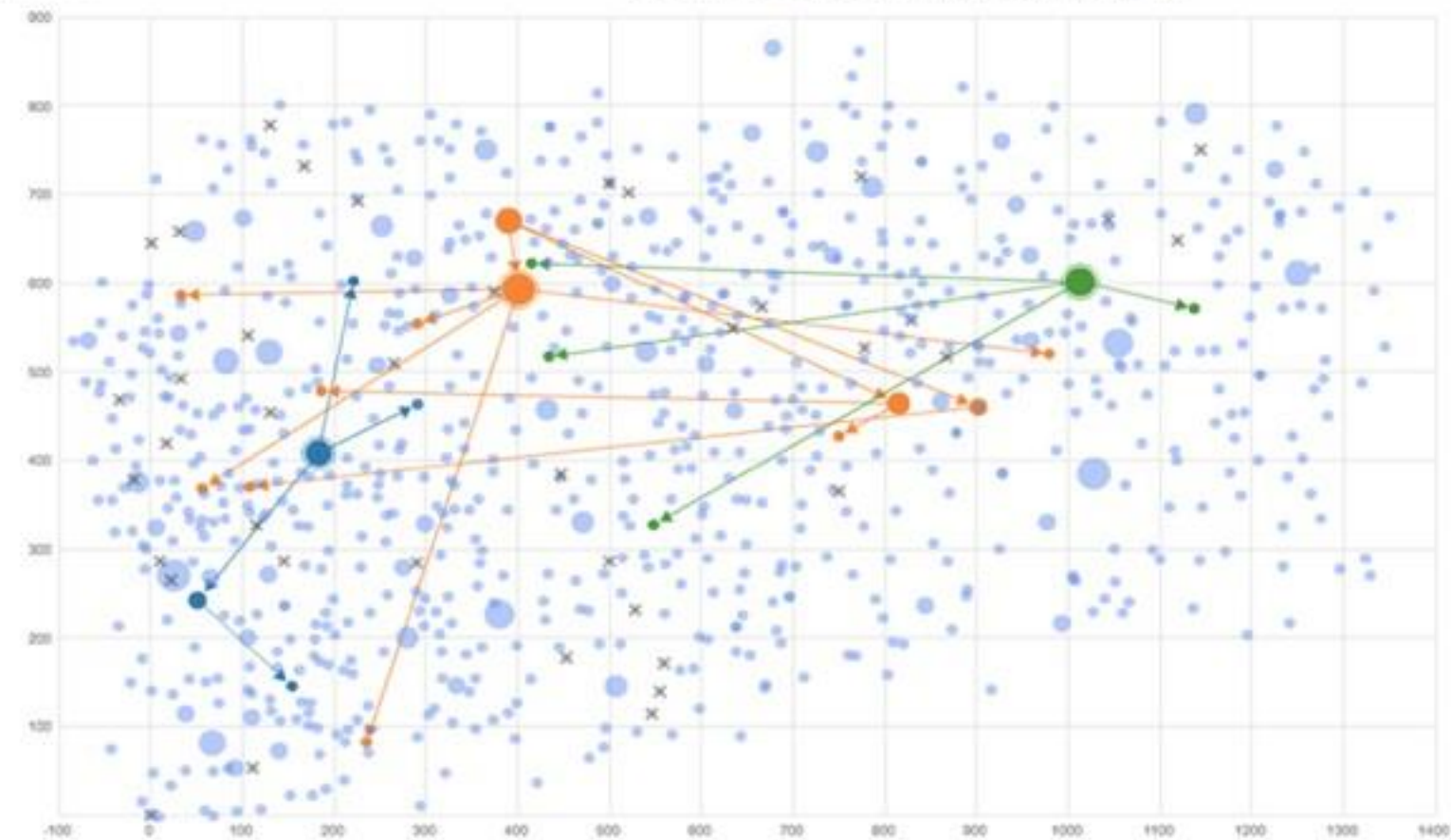
E.J. Marey's sphygmograph [from Braun 83]



[illegible]



Colony Viewer - colony.observations.txt | colony.ancestry.txt



id	E	N	first	last
1	808	413	2002	2004
2	399	126	2002	2003
3	333	148	2002	2008
4	56	333	2002	2003
5	9	543	2002	2008
6	141	451	2002	2008
7	434	517	2003	2008
8	1253	575	2003	2008
9	1160	442	2003	2004
10	758	575	2003	2008
11	435	776	2003	2008
12	-54	486	2003	2008
13	799	545	2003	2008
14	415	622	2003	2008
15	593	428	2003	2008
16	989	682	2003	2008
17	613	703	2003	2008
18	290	252	2003	2008
20	360	771	2004	2008
21	623	694	2004	2005
22	879	520	2004	2008
23	1144	523	2004	2008
24	584	312	2004	2008
25	463	264	2004	2008
26	597	673	2004	2008
27	425	738	2004	2008
28	431	661	2004	2008
29	853	306	2004	2008
30	130	130	1988	2008
31	270	565	1988	2008
32	326	483	2004	2008
33	255	552	1988	2008
34	260	566	2004	2008

Colony Viewer



1995 - 2009

2 Years Gone

First Seen Only

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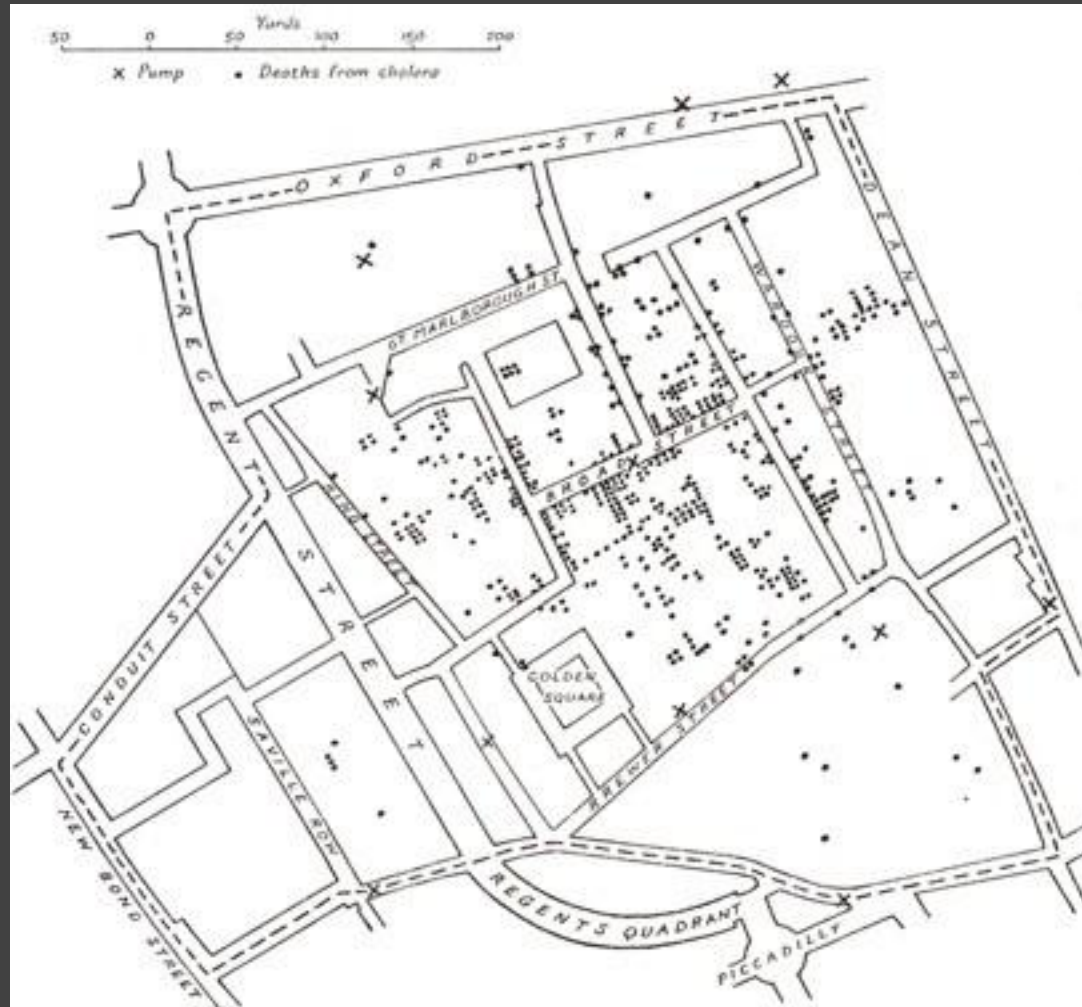
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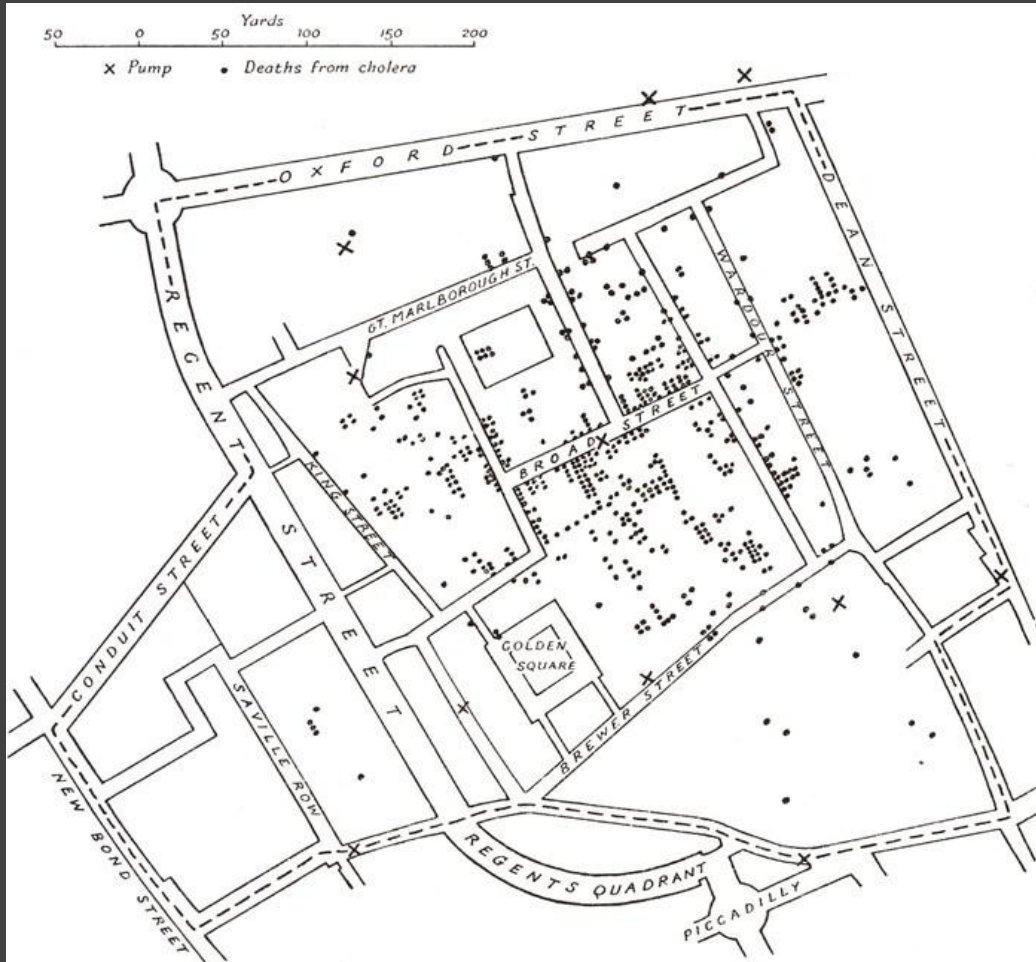
Collaborate and revise

Data in context: Cholera outbreak



In 1854 John Snow plotted the position of each cholera case on a map. [from Tufte 83]

Data in context: Cholera outbreak



Used map to hypothesize that pump on Broad St. was the cause. [from Tufte 83]

Make a decision: Challenger

HISTORY OF O-RING DAMAGE ON SRM FIELD JOINTS

	SRM No.	Cross Sectional View			Top View		Clocking Location (deg)
		Erosion Depth (in.)	Perimeter Affected (deg)	Nominal Dia. (in.)	Length Of Max Erosion (in.)	Total Heat Affected Length (in.)	
61A LH Center Field**	22A	None	None	0.280	None	None	36° - 66°
61A LH Center Field**	22A	NONE	NONE	0.280	NONE	NONE	338° - 18°
51C LH Forward Field**	15A	0.010	154.0	0.280	4.25	5.25	163
51C RH Center Field (prim)***	15B	0.038	130.0	0.280	12.50	58.75	354
51C RH Center Field (sec)***	15B	None	45.0	0.280	None	29.50	354
41D RH Forward Field	13B	0.028	110.0	0.280	3.00	None	275
41C LH Aft Field*	11A	None	None	0.280	None	None	--
41B LH Forward Field	10A	0.040	217.0	0.280	3.00	14.50	351
STS-2 RH Aft Field	2B	0.053	116.0	0.280	--	--	90

*Hot gas path detected in putty. Indication of heat on O-ring, but no damage.
 **Soot behind primary O-ring.
 ***Soot behind primary O-ring, heat affected secondary O-ring.

Clocking location of leak check port - 0 deg.

OTHER SRM-15 FIELD JOINTS HAD NO BLOWHOLES IN PUTTY AND NO SOOT NEAR OR BEYOND THE PRIMARY O-RING.

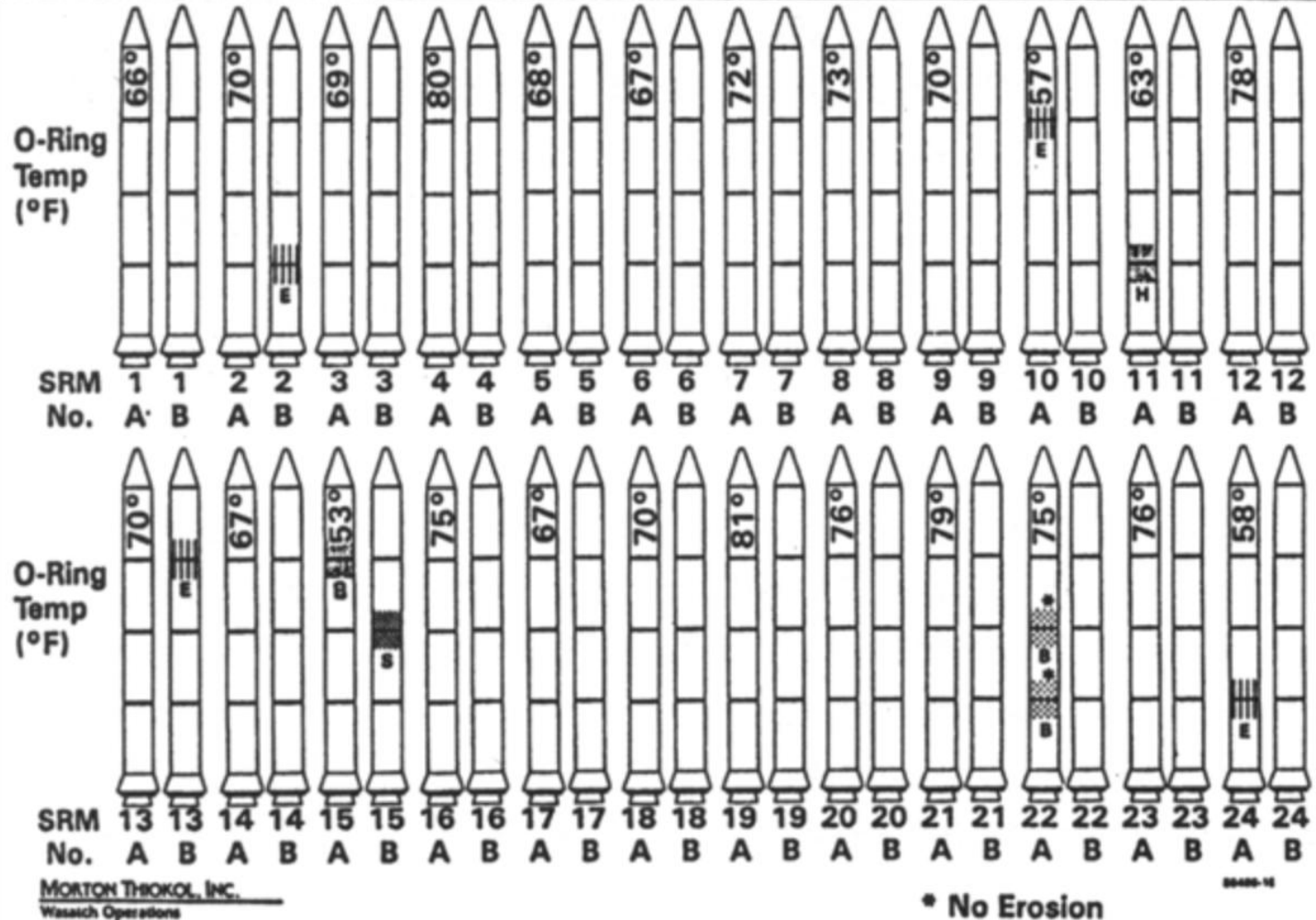
SRM-22 FORWARD FIELD JOINT HAD PUTTY PATH TO PRIMARY O-RING, BUT NO O-RING EROSION AND NO SOOT BLOWBY. OTHER SRM-22 FIELD JOINTS HAD NO BLOWHOLES IN PUTTY.

BLOW BY HISTORY		HISTORY OF O-RING TEMPERATURES (DEGREES - F)				
SRM-15 WORST BLOW-BY		MOTOR	MBT	AMB	O-RING	WIND
o 2 CASE JOINTS (80°), (110°) ARC		DM-1	68	36	47	10 MPH
o MUCH WORSE VISUALLY THAN SRM-22		DM-2	76	45	52	10 MPH
SRM 22 BLOW-BY		QM-3	72.5	40	48	10 MPH
o 2 CASE JOINTS (30-40°)		QM-4	76	48	51	10 MPH
SRM-13A, 15, 16A, 18, 23A 24A		SRM-15	52	64	53	10 MPH
o NOZZLE BLOW-BY		SRM-22	77	78	75	10 MPH
		SRM-25	55	26	29	10 MPH
					27	25 MPH

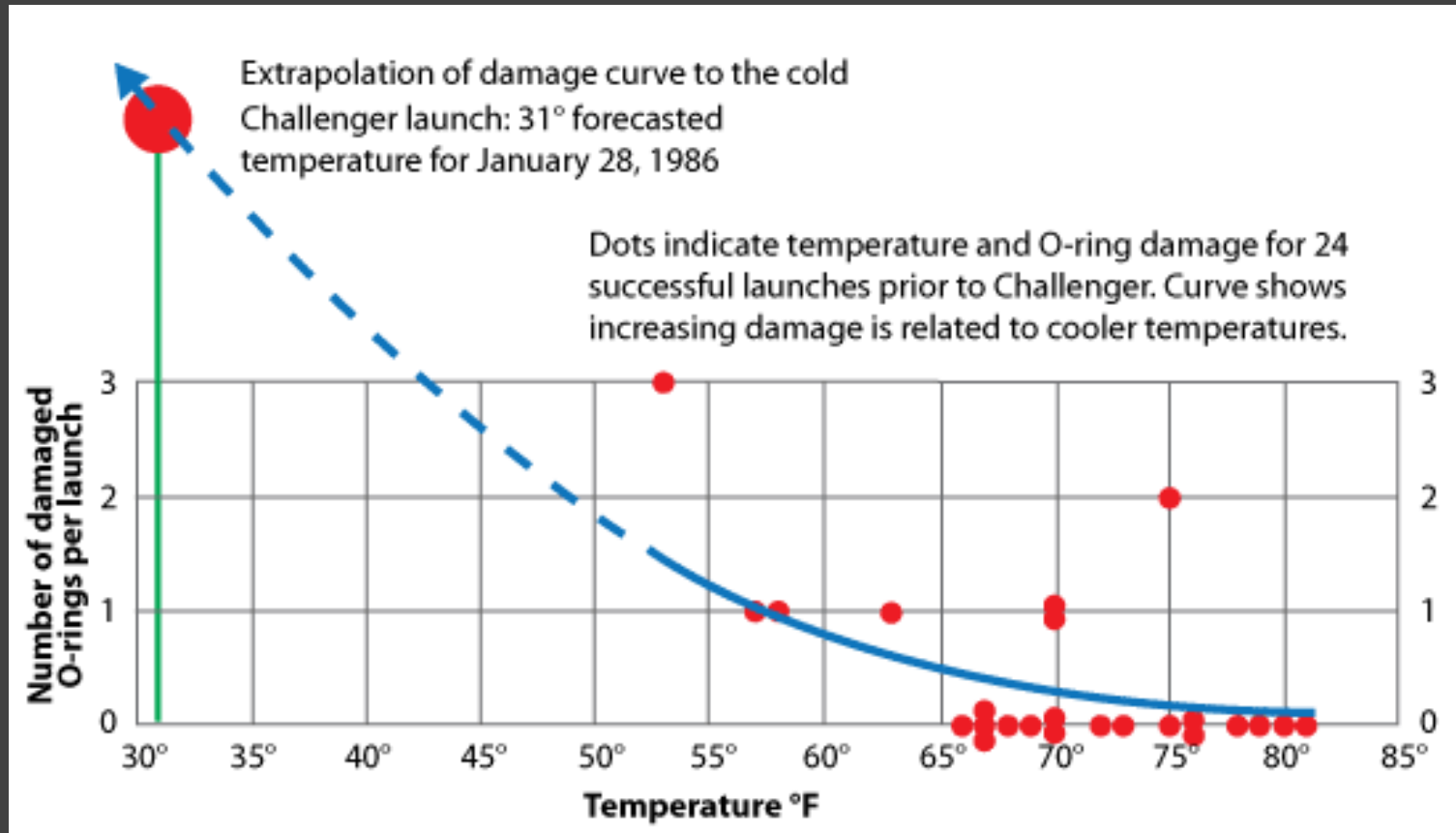
2 of 13 pages of material faxed to NASA by Morton Thiokol [from Tufte 1997]

Make a decision: Challenger

History of O-Ring Damage in Field Joints (Cont)



Make a decision: Challenger



Visualizations drawn by Tufte show how low temperatures damage O-rings [Tufte 97]

HW2 Times (previous class)

8 17 16 3 10 6 4 10 7 8 17 11 8 6 7 6 6 8 14 5 7 8 4 9 6 3
10 6 8 8 6 6 7 7 7 12 10 10 6 8 12 8 7 6 12 7 9 56 10 48 12
12 13 5 8 12 9 8 6 14 12 5 7 7 10 10 5 15 12 10 16 8 3 20
10 6 5 15 12 9 11 24 5 7 7 8 8 10 10 5 25 8 10 4 8 10 7 13
7 5 8 12 11 10 10 17 10 24 10 4 10 12 60 6 5 9 13 24 20 8
15 6 8 7 9 11 6 5 1 5 6 12 6 6 60 12 11 20 75 5 24 11 6 12
12 8 20 14 19 6 10 12 9 8 24 9

Why Create Visualizations?

1. **Record** information

Blueprints, photographs, seismographs, ...

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Find patterns

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Set A

X	Y
10	8.04
8	6.95
13	7.58
9	8.81
11	8.33
14	9.96
6	7.24
4	4.26
12	10.84
7	4.82
5	5.68

Set B

X	Y
10	9.14
8	8.14
13	8.74
9	8.77
11	9.26
14	8.1
6	6.13
4	3.1
12	9.11
7	7.26
5	4.74

Set C

X	Y
10	7.46
8	6.77
13	12.74
9	7.11
11	7.81
14	8.84
6	6.08
4	5.39
12	8.15
7	6.42
5	5.73

Set D

X	Y
8	6.58
8	5.76
8	7.71
8	8.84
8	8.47
8	7.04
8	5.25
19	12.5
8	5.56
8	7.91
8	6.89

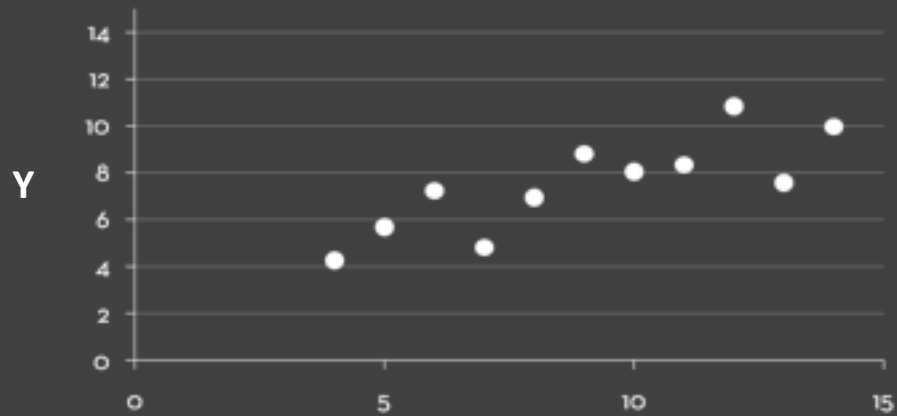
Summary Statistics

$$u_x = 9.0 \quad \sigma_x = 3.317$$

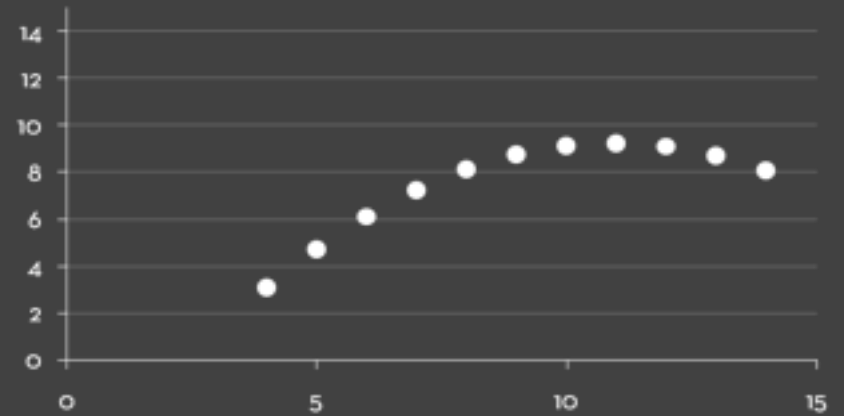
$$u_y = 7.5 \quad \sigma_y = 2.03$$

[Anscombe 73]

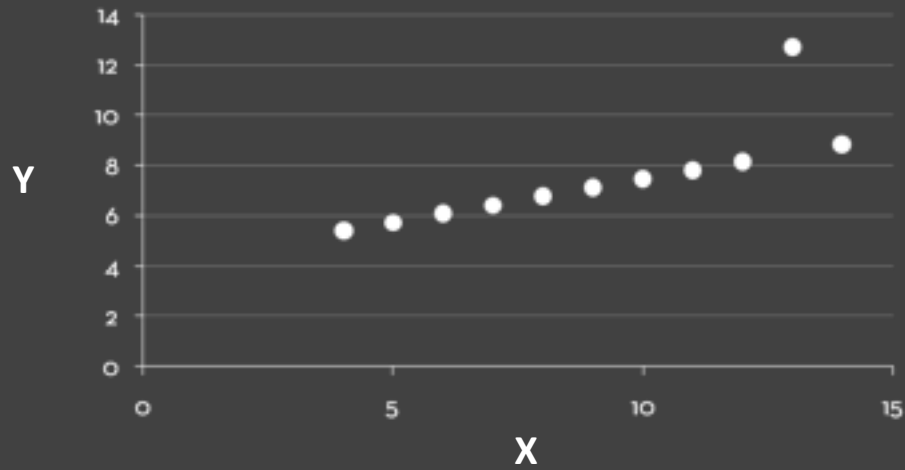
Set A



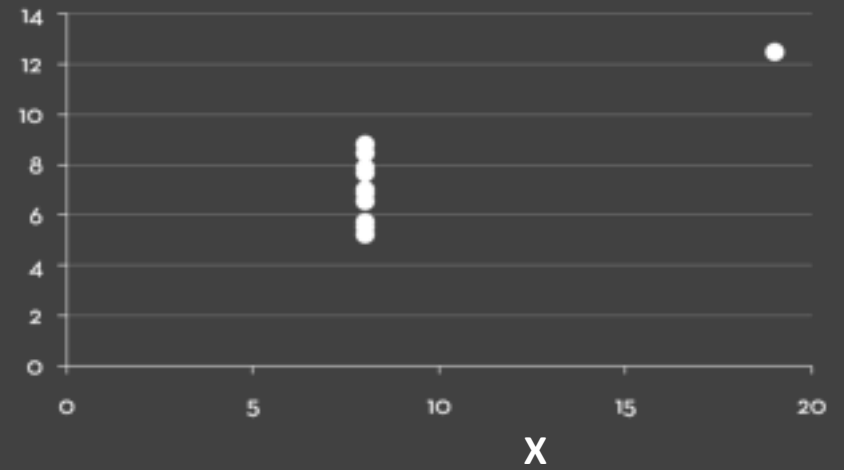
Set B



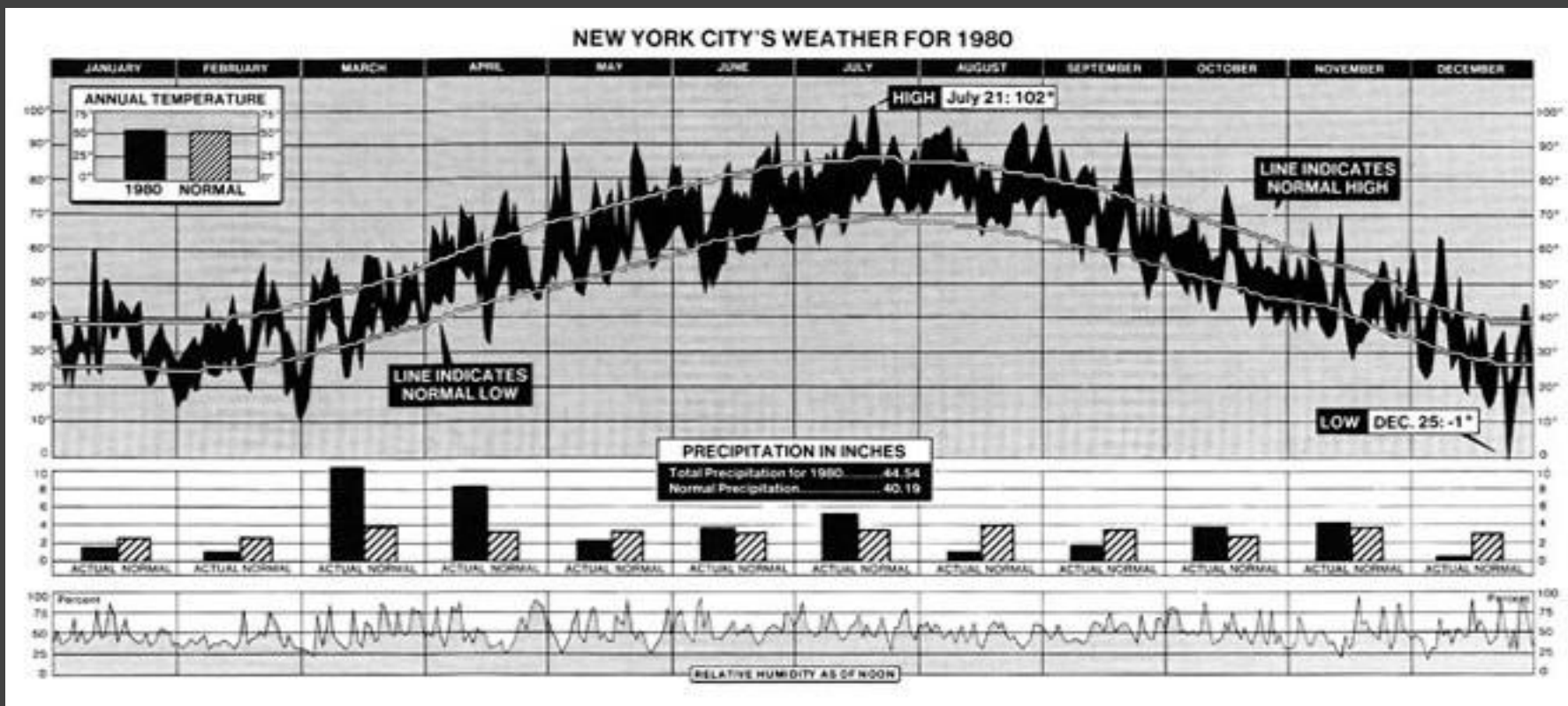
Set C



Set D



Find patterns: NYC weather



From the New York Times 1981

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How to gauge the quality of a visualization?

“The first sign that a visualization is good is that it shows you **a problem in your data**...

...every successful visualization that I've been involved with has had this stage where you realize, "Oh my God, this data is not what I thought it would be!" So already, you've discovered something.”

- Martin Wattenberg

Age: 95
Sex: Female
Race: Caucasian
County (Res): Prince Georges
Zip Code (Res): 20770
Received: 940706
Complaint Sequence: 1
Source: Citizen
Reason: Delinquent
Alleged Offense: HARAS
Offense Level: 2 - Misdemeanor
County (Off): Prince Georges
Zip Code (Off): 20770
Area: V
Office: 71610
Intake Decision Date: 940729
Intake Decision: Closed
Days to ID: 23
Court Finding: NONE
Disposition Date: 0
Disposition:

Offens

Count

Area:

Office:

Intake

TC

Query Result: 4792 out of 4792 (100%)



Graph Viewer

Roll-up by:

All

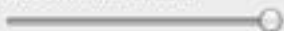
Visualization:

Node-Link

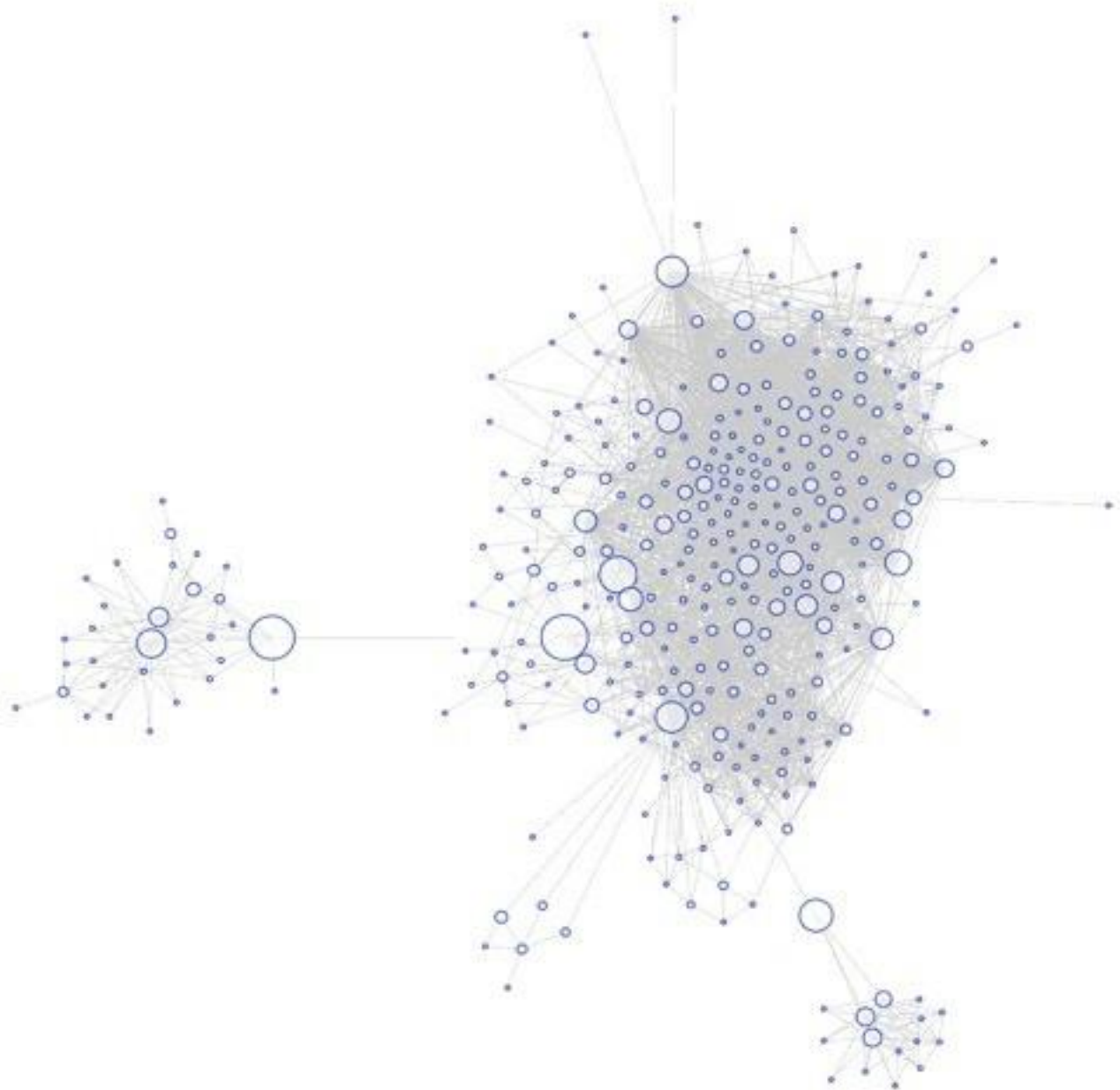
Sort by:

None

Edge centrality filters:



- ☐ Images
- ☒ Animate



Graph Viewer

Roll-up by:

All

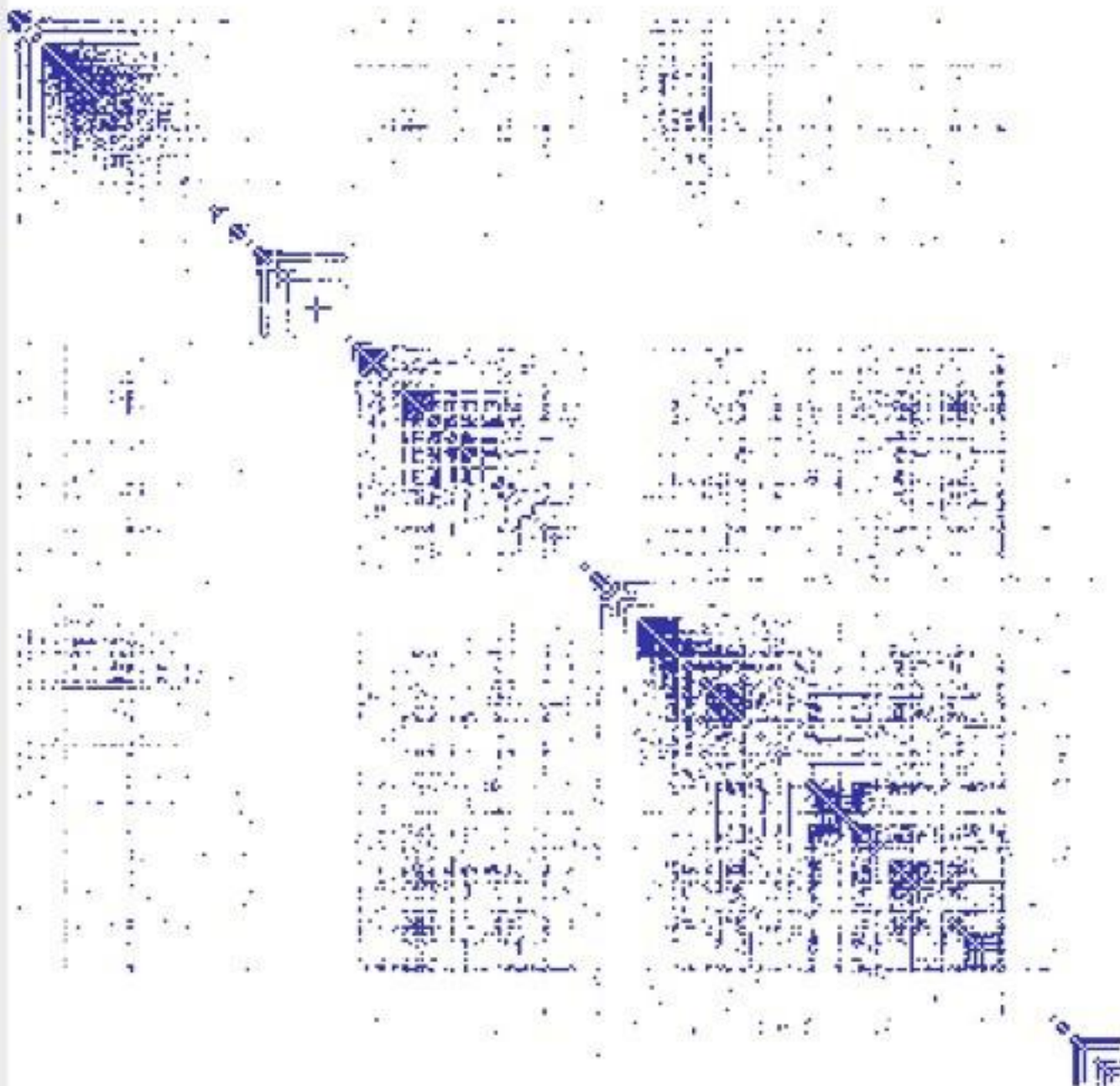
Visualization:

Matrix

Sort by:

Linkage

Edge centrality filters:



Graph Viewer

Roll-up by:

All

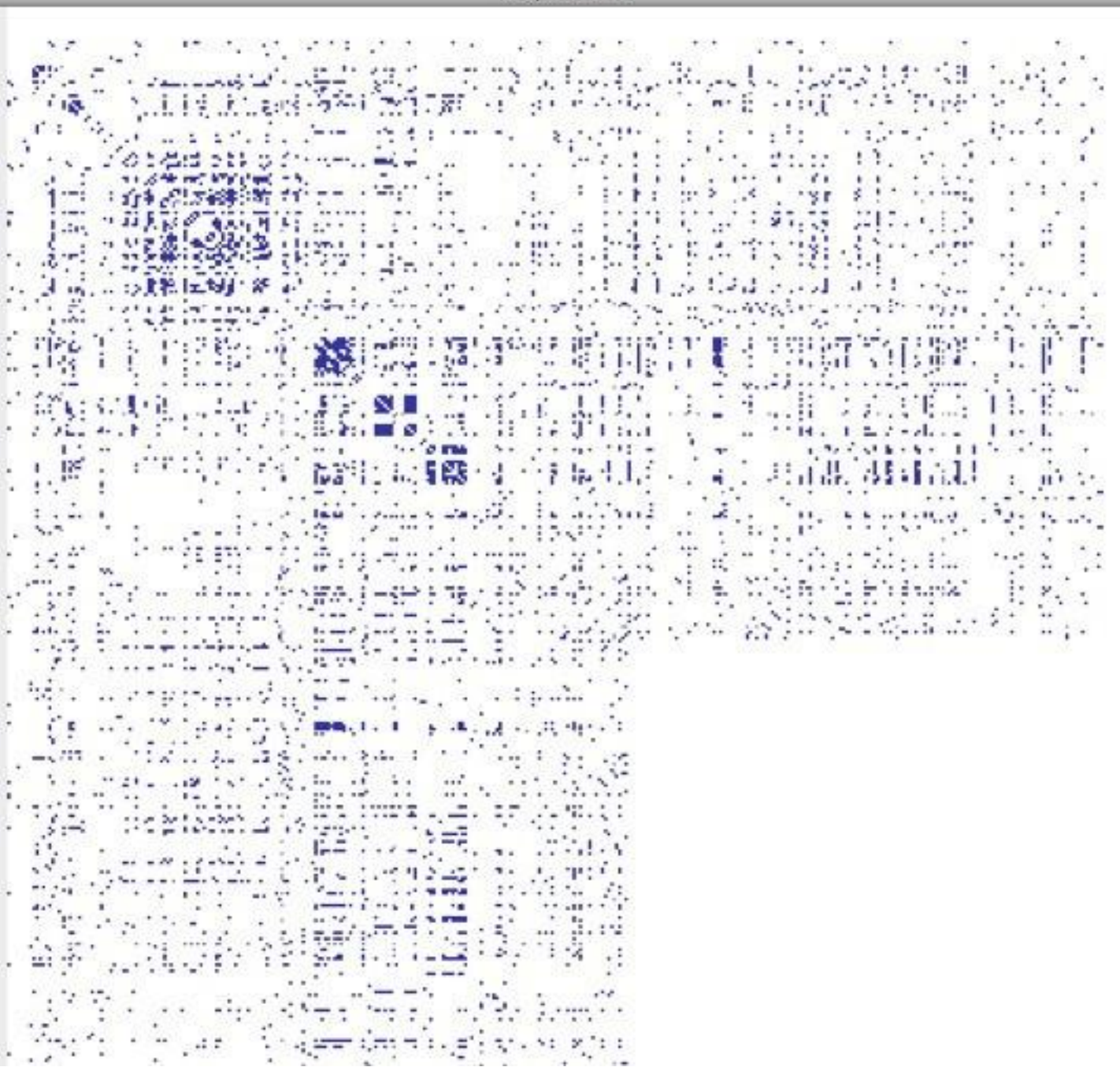
Visualization:

Matrix

Sort by:

None

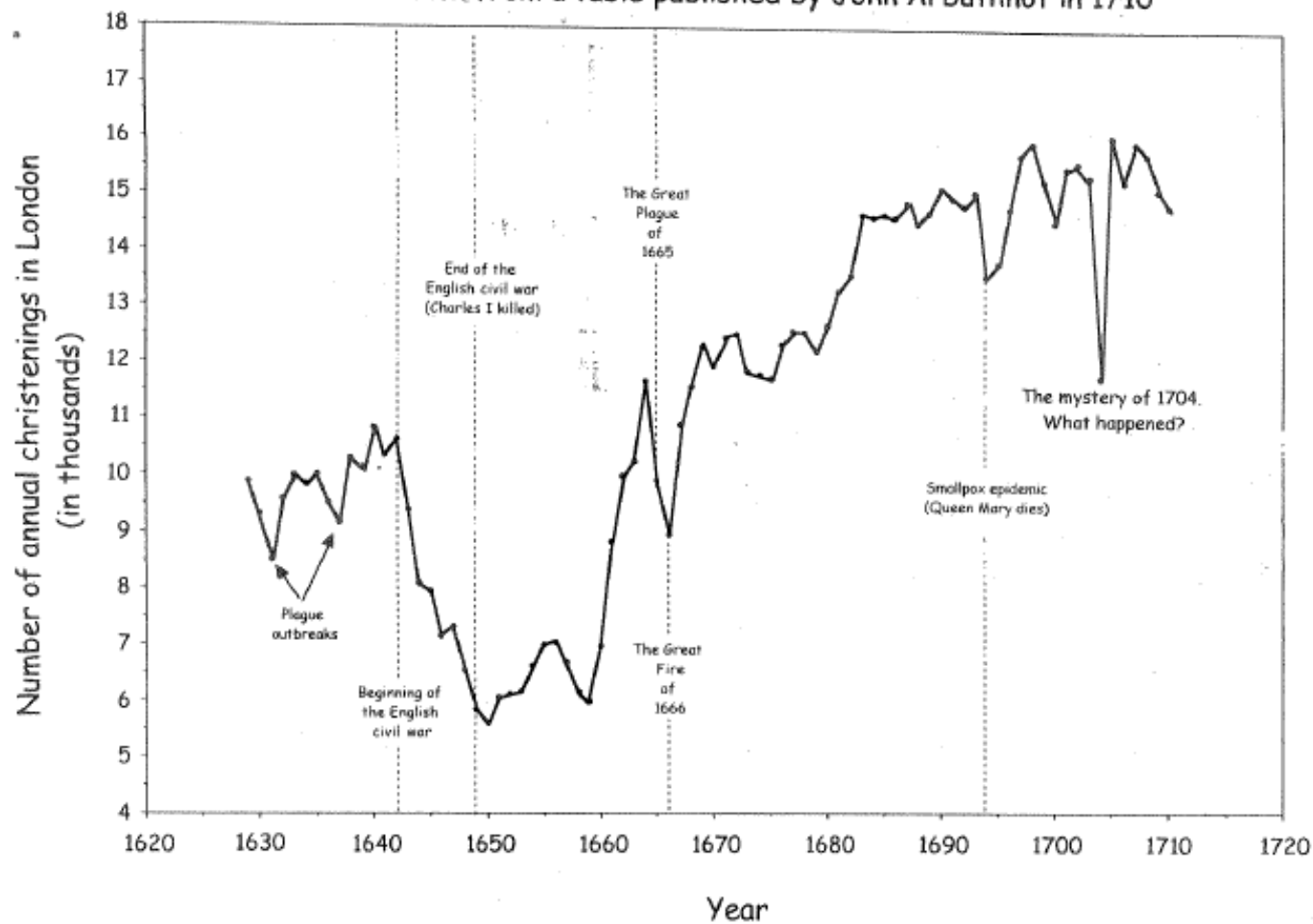
Edge centrality filters:



Visualize Friends by School?

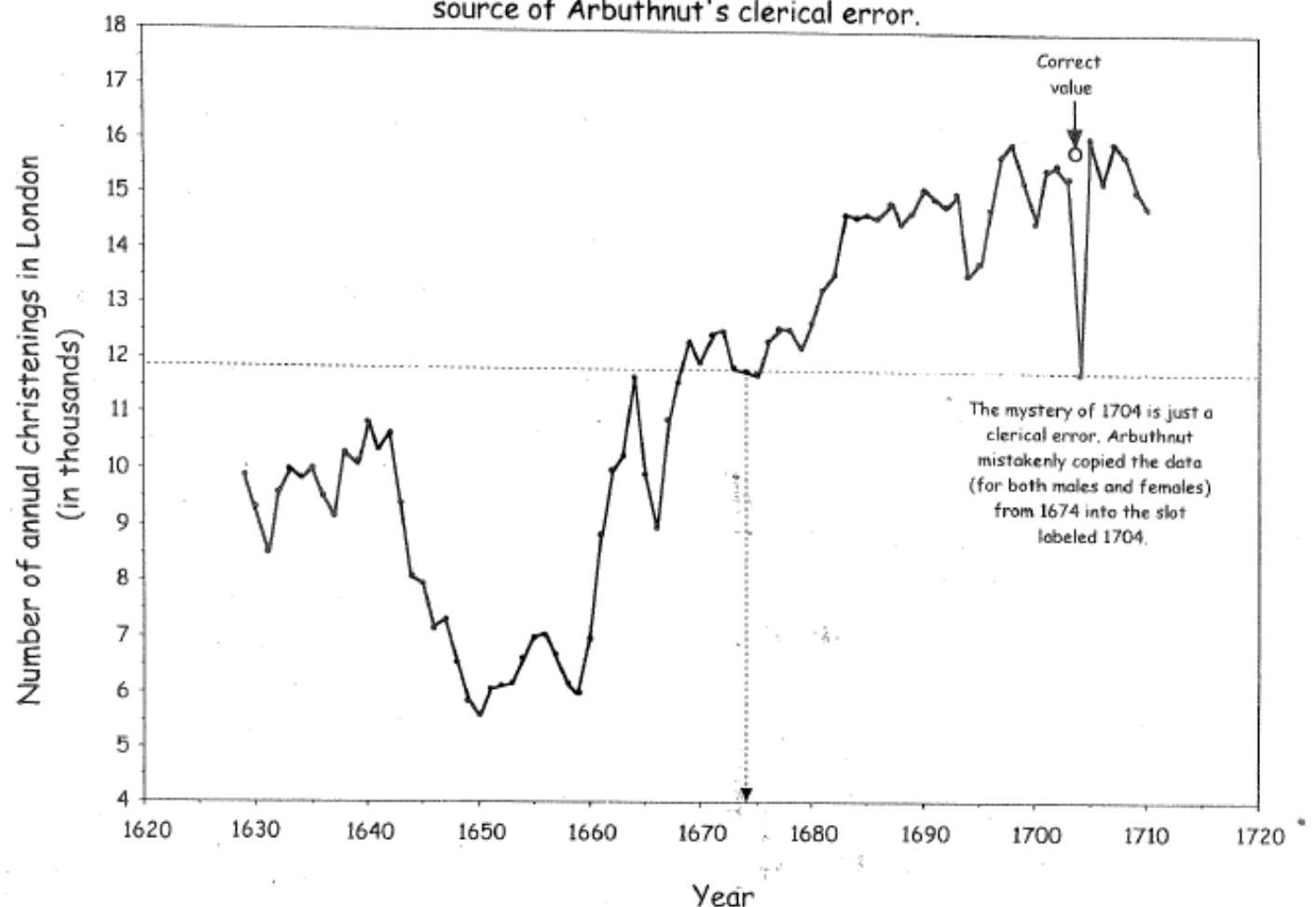


A plot of the annual christenings in London from the London Bills of Mortality.
 These data were taken from a table published by John Arbuthnot in 1710



Date source: Arbuthnot, 1710

The mystery of 1704 is resolved by noting that there was only one other point (1674) with exactly the same values as the 1704 outlier. This proved to be the source of Arbuthnot's clerical error.



Data source: Arbuthnot, 1710

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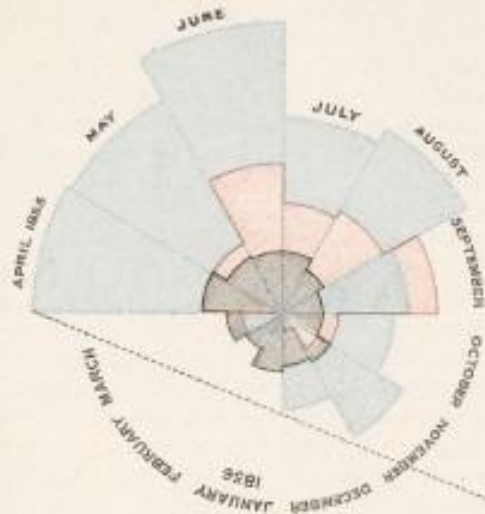
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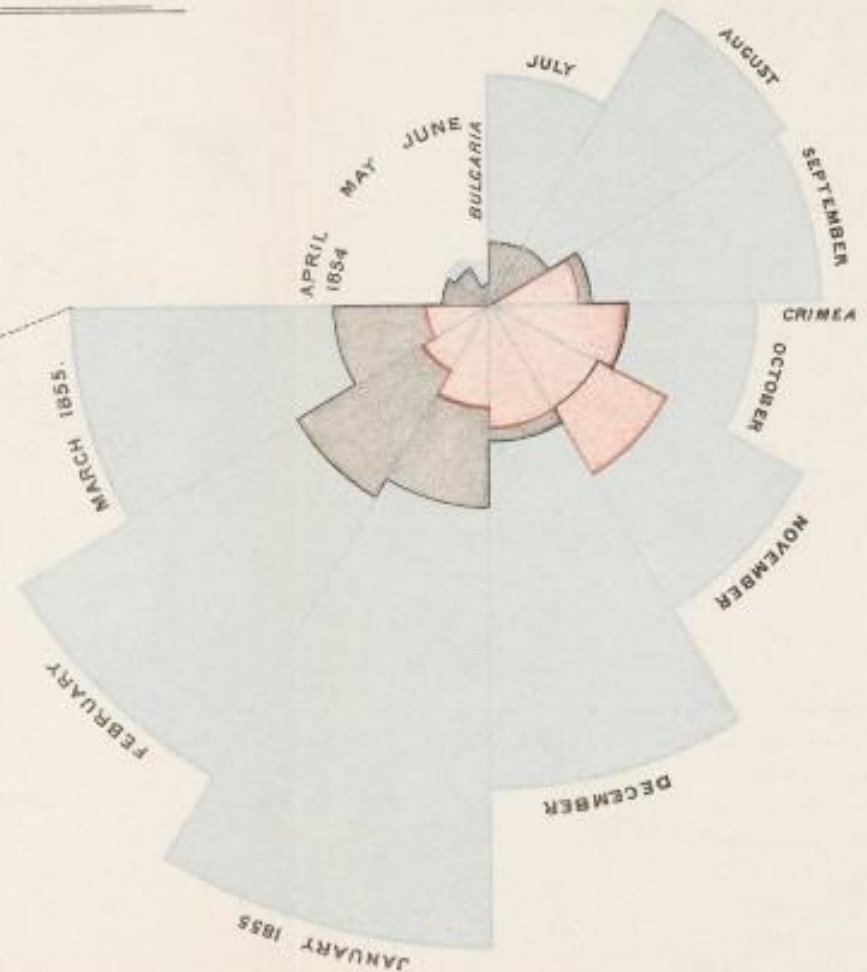
Collaborate and revise

DIAGRAM OF THE CAUSES OF MORTALITY IN THE ARMY IN THE EAST.

2.
APRIL 1855 to MARCH 1856.



1.
APRIL 1854 to MARCH 1855.



The Areas of the Blue, red, & black wedges are each measured from

“to affect thro’ the eyes
what we fail to convey to
the public through their
word-proof ears”

Harrison & Sons, R. Martineau Lane

1856 “Coxcomb” of Crimean War Deaths, Florence Nightingale

Things to keep in mind when visualizing data

SIX PRINCIPLES

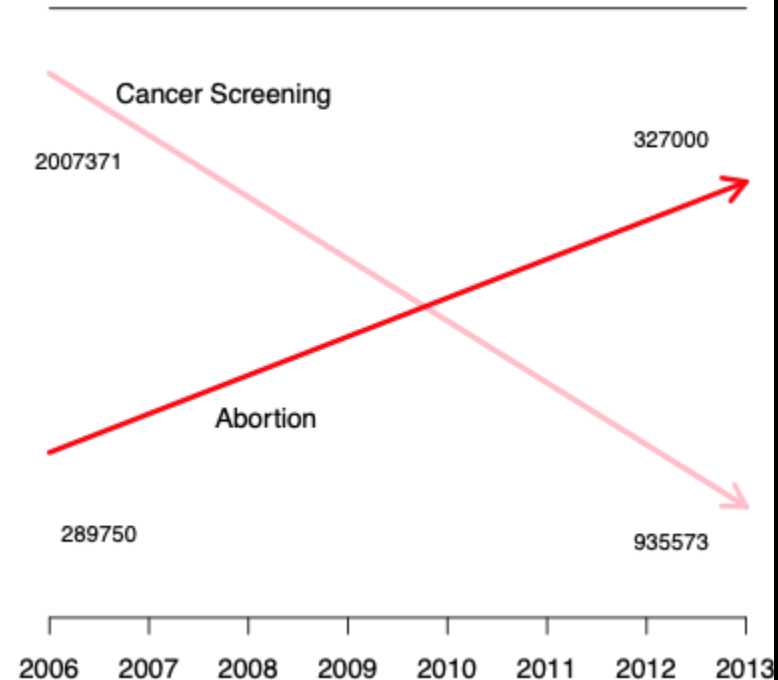
- 1. Scale**
- 2. Conditioning**
- 3. Perception**
- 4. Transformations**
- 5. Context**
- 6. Smoothing**

PRINCIPLES OF SCALE

Choose axes well

CASE 1: PLANNED PARENTHOOD 2015 HEARING

- Investigation of federal funding of Planned Parenthood in light of fetal tissue controversy
- Congressman Chaffetz (R-UT) showed plot which originally appeared in a report by Americans United for Life (<http://www.aul.org/>)

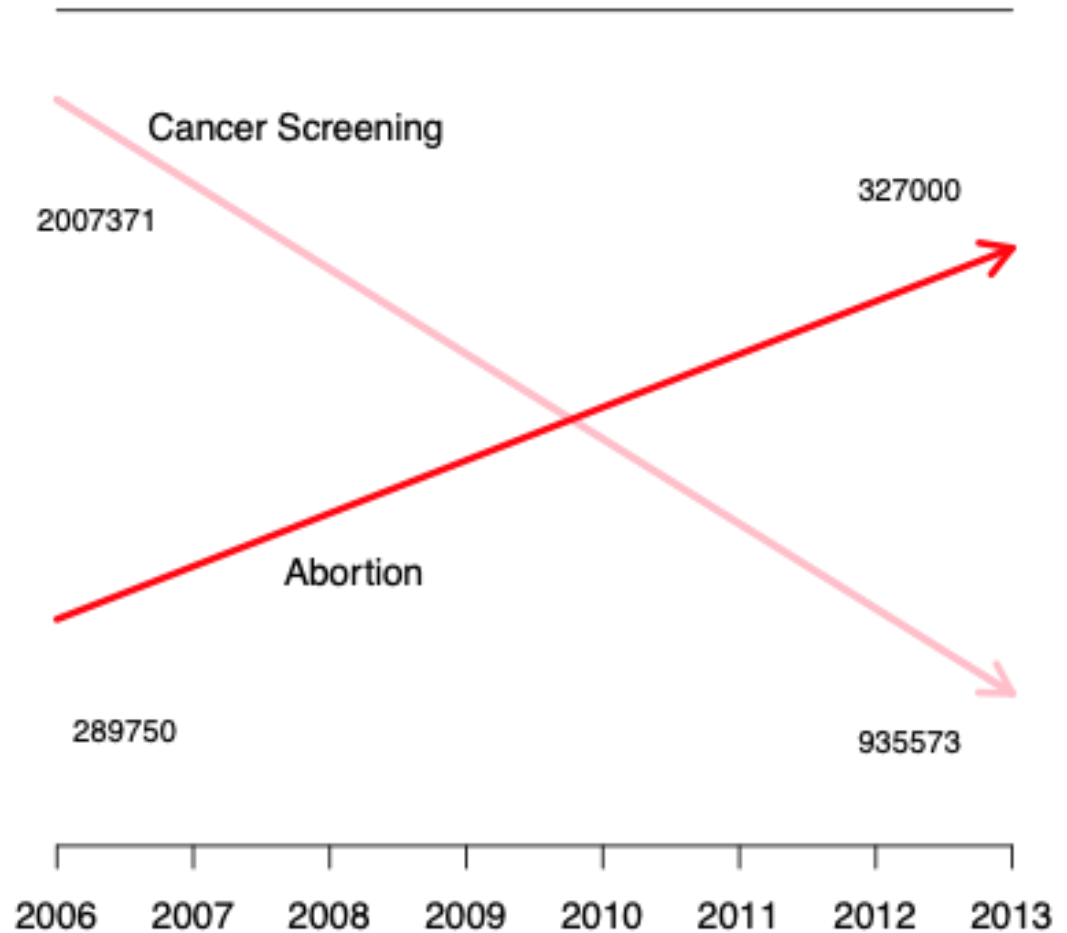


Full Report available at

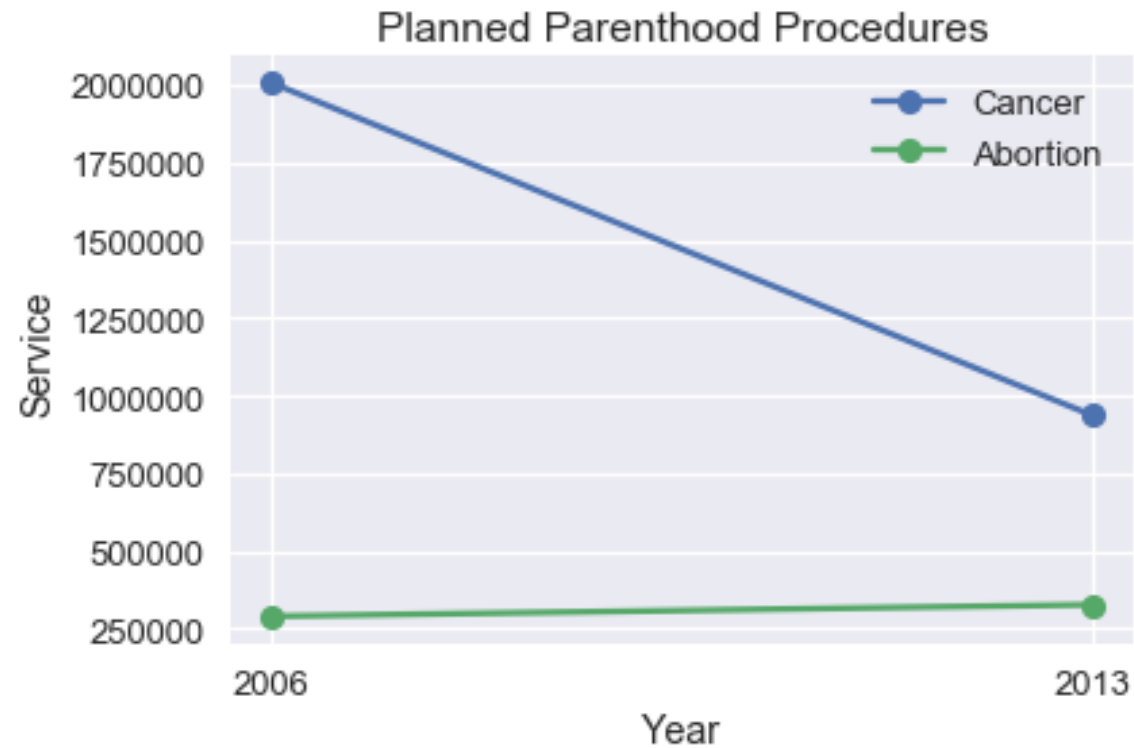
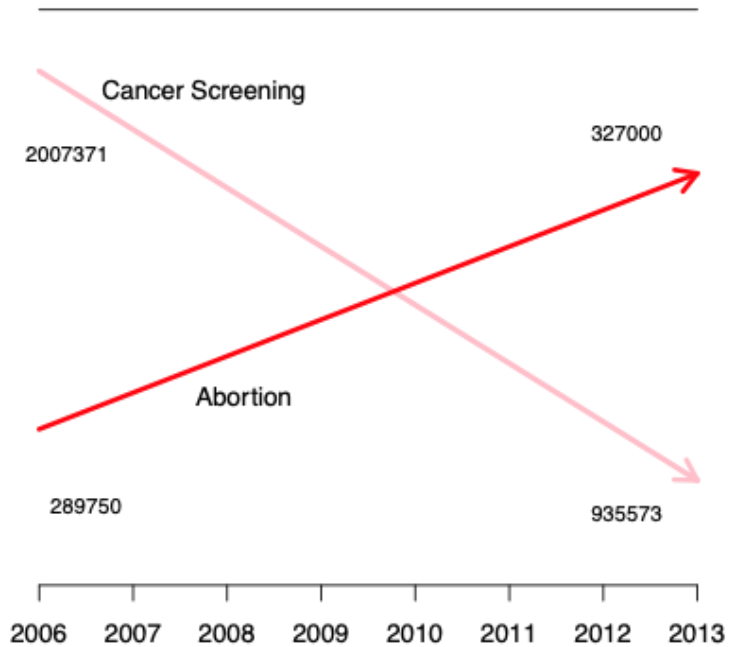
<https://oversight.house.gov/interactivepage/plannedparenthood/> .

CASE 1: PLANNED PARENTHOOD 2015 HEARING

- Procedures: cancer screenings and abortions
- What message is this plot trying to convey?
- What is suspicious?
- How many data points are plotted?

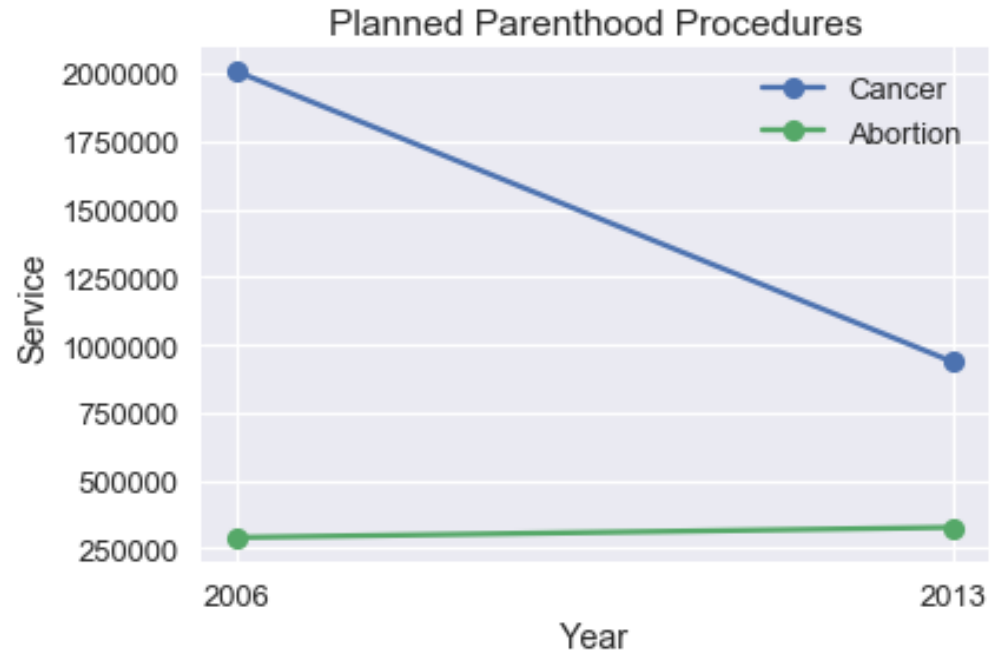


SCALE



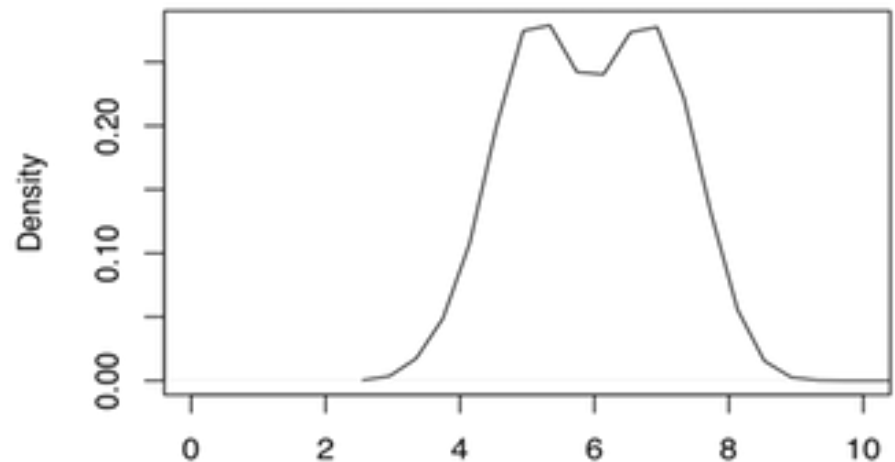
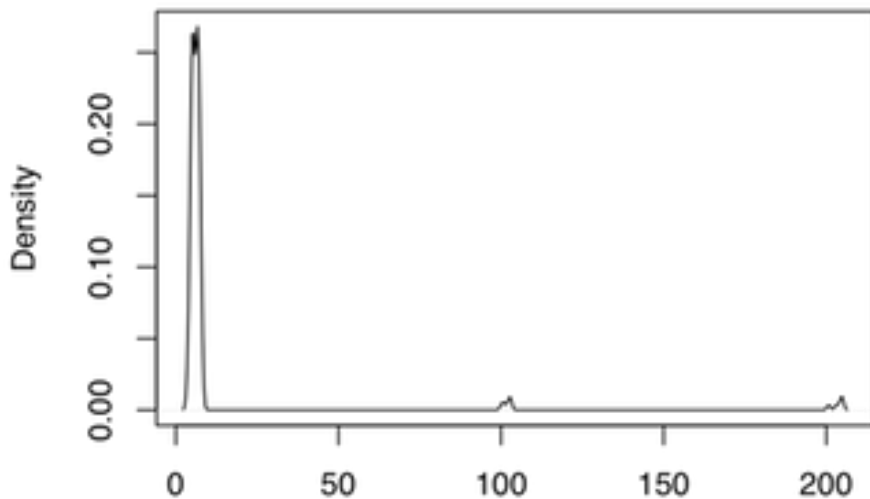
KEEP CONSISTENT AXIS SCALES

- Don't change scale mid-axis
- Don't use two different scales for same axis
- How does this plot change perception of information?



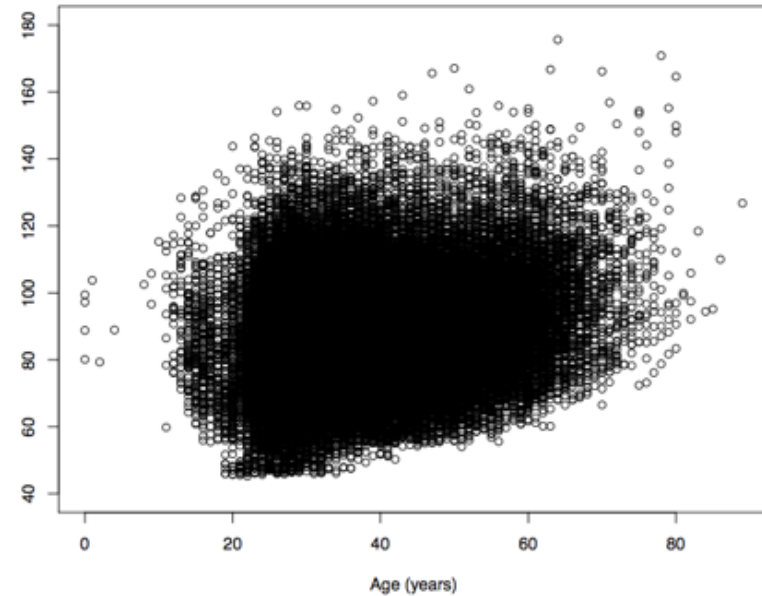
SCALE: REVEAL THE DATA

- Choose axis limits to fill plot
- If necessary, zoom into region with most of data
 - Can make separate plots for different regions



CHERRY BLOSSOM RUNNERS

- 10 mi run in DC every April
- Results available from 1999-2019
- In 2019 over 17,000 runners
- Scatter plot of run time (min) against age (yrs)
- What's the relationship between run time and age?



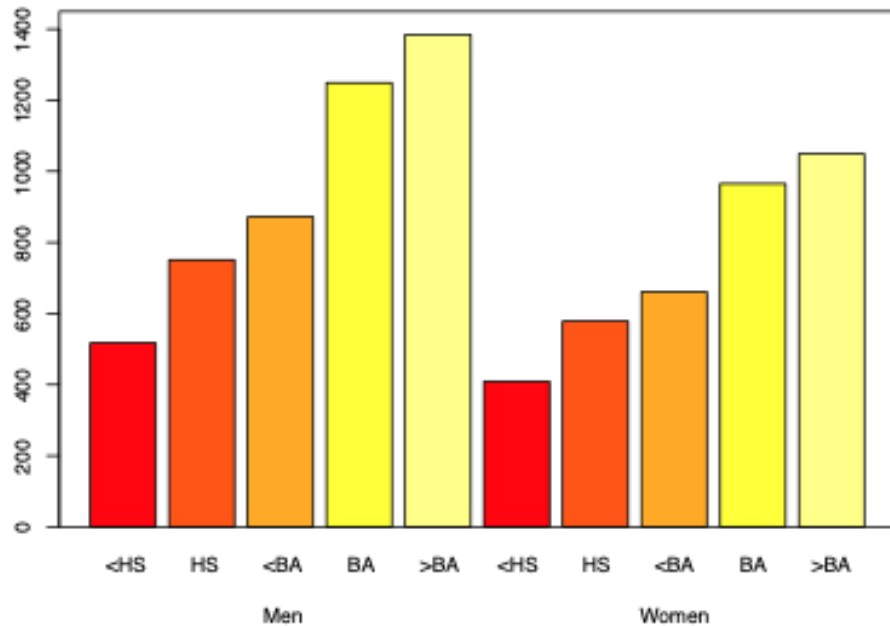
<http://www.cherryblossom.org/>

PRINCIPLES OF CONDITIONING

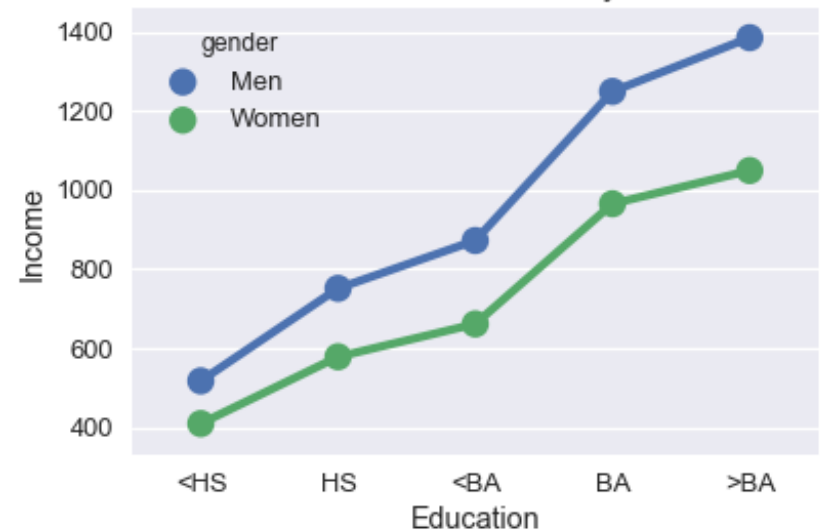
Show distributions and relationships between
subgroups of our data

CONDITIONING

2014 Median Weekly Earnings
Full-Time Workers over 25 years old



2014 Median Weekly Earnings
Full-Time Workers over 25 years old



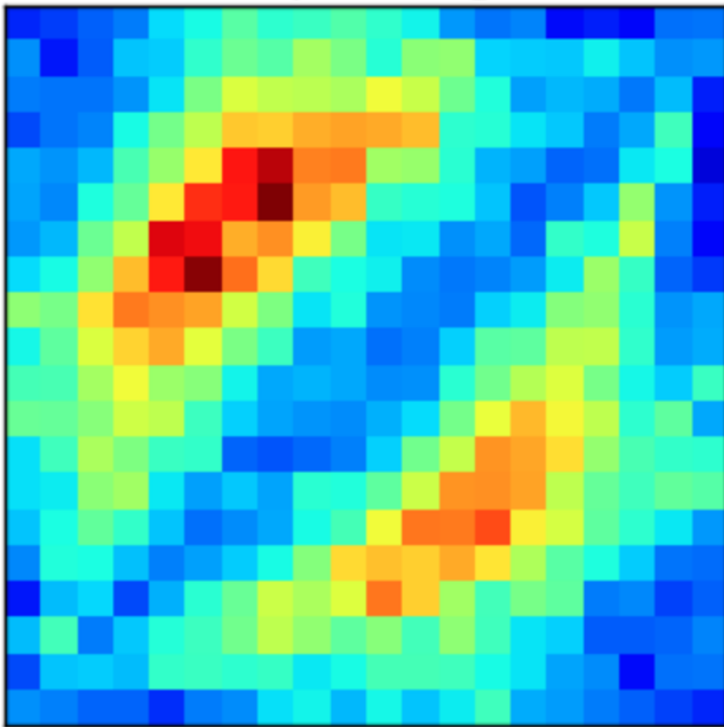
- **Conditioning on male/female aligns points on x-axis**
 - What does it reveal?
 - Why is this interesting?

PRINCIPLES OF PERCEPTION

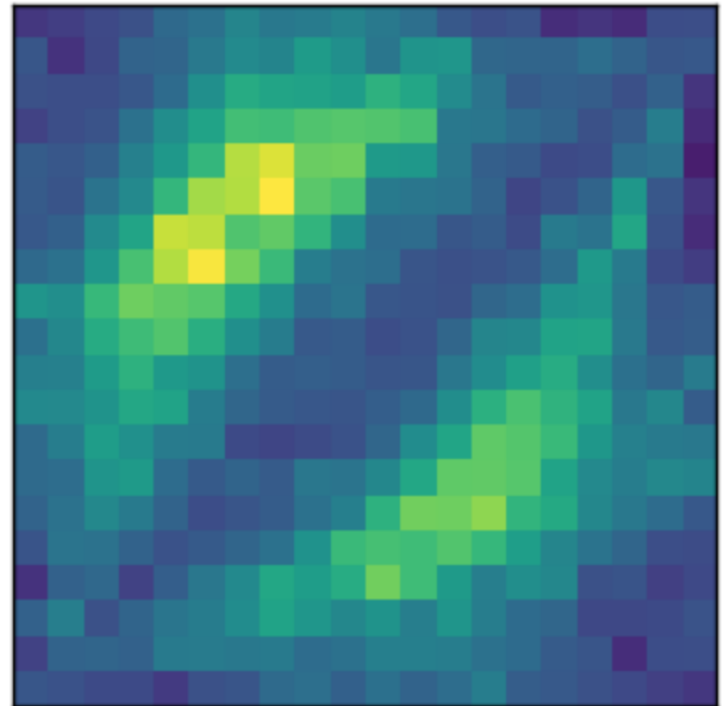
Human perception has specific properties that
are important to consider in visualization
design

COLOR CHOICES MATTER!

JET COLORMAP



VIRIDIS COLORMAP



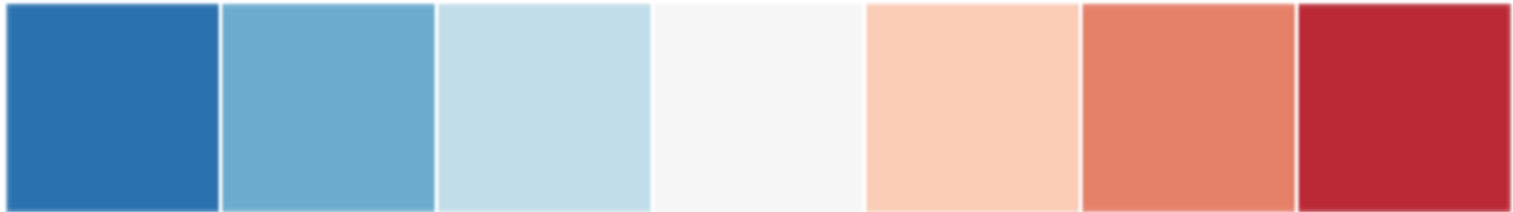
USE A PERCEPTUALLY UNIFORM COLOR MAP

- **Perceptually uniform: changing data from 0.1 to 0.2 appears similar to change from 0.8 to 0.9.**
 - Measure by running experiments on people!
- **Jet, the old matplotlib default, was far from uniform!**
- **Also, avoid red + green since many people are colorblind**
- **More details, if interested:**
<https://bids.github.io/colormap/>

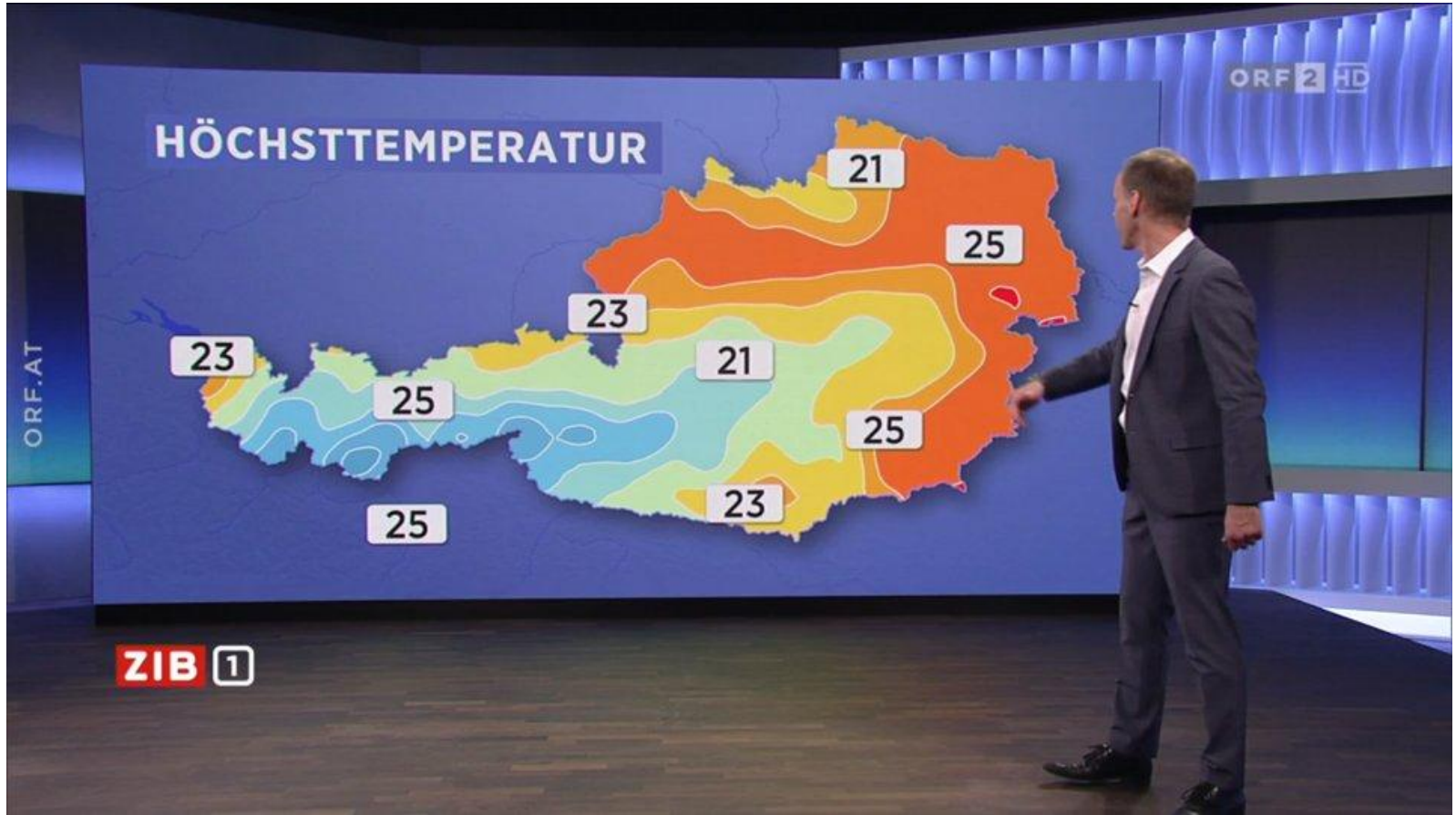
USE COLOR TO HIGHLIGHT DATA TYPE

- Do both low and high value deserve equal emphasis? Use a diverging scheme where light colors represent middle values

```
sns.palettes(sns.color_palette("RdBu_r", 7))
```



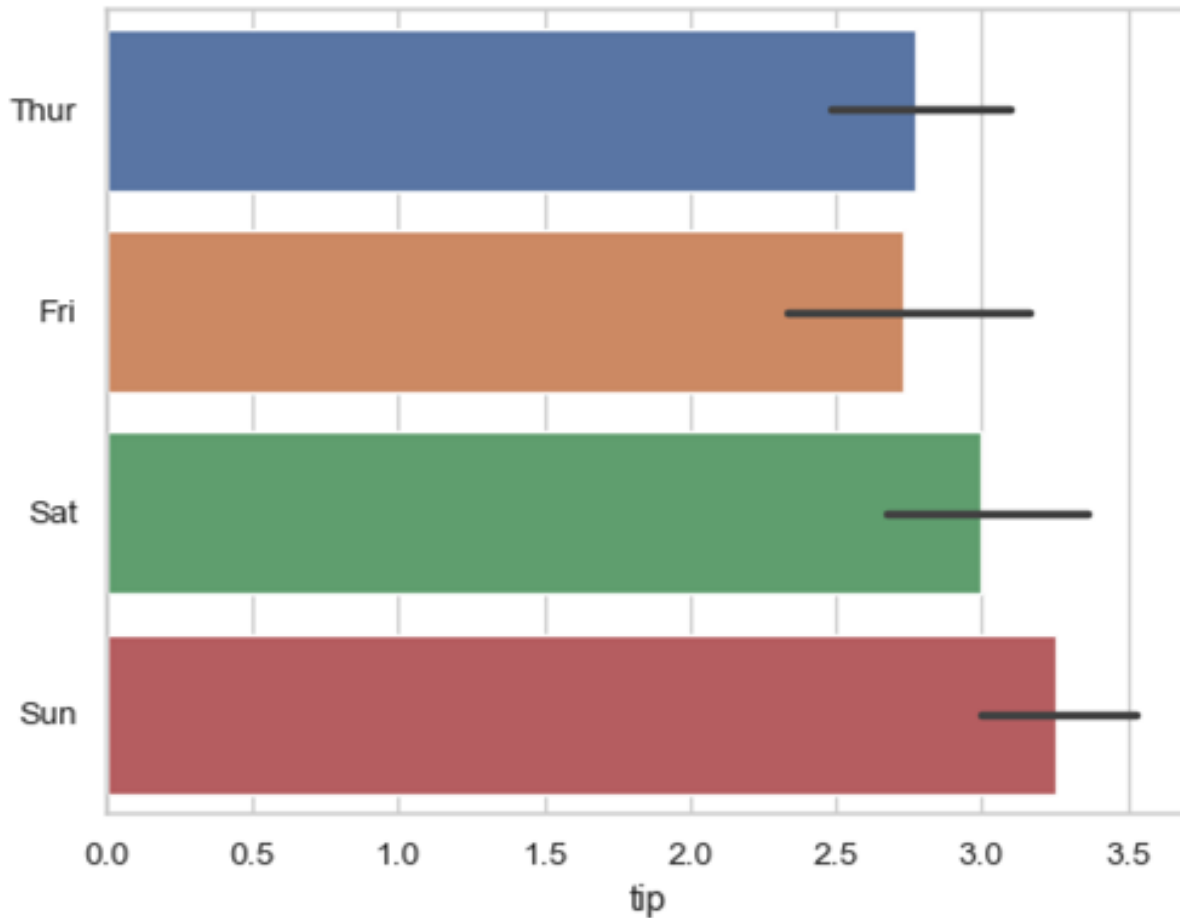
BONUS: WHAT COLOR IS 25 HERE?



LENGTHS ARE EASY TO UNDERSTAND

People can easily distinguish two different lengths

E.g. Heights of bars in bar chart

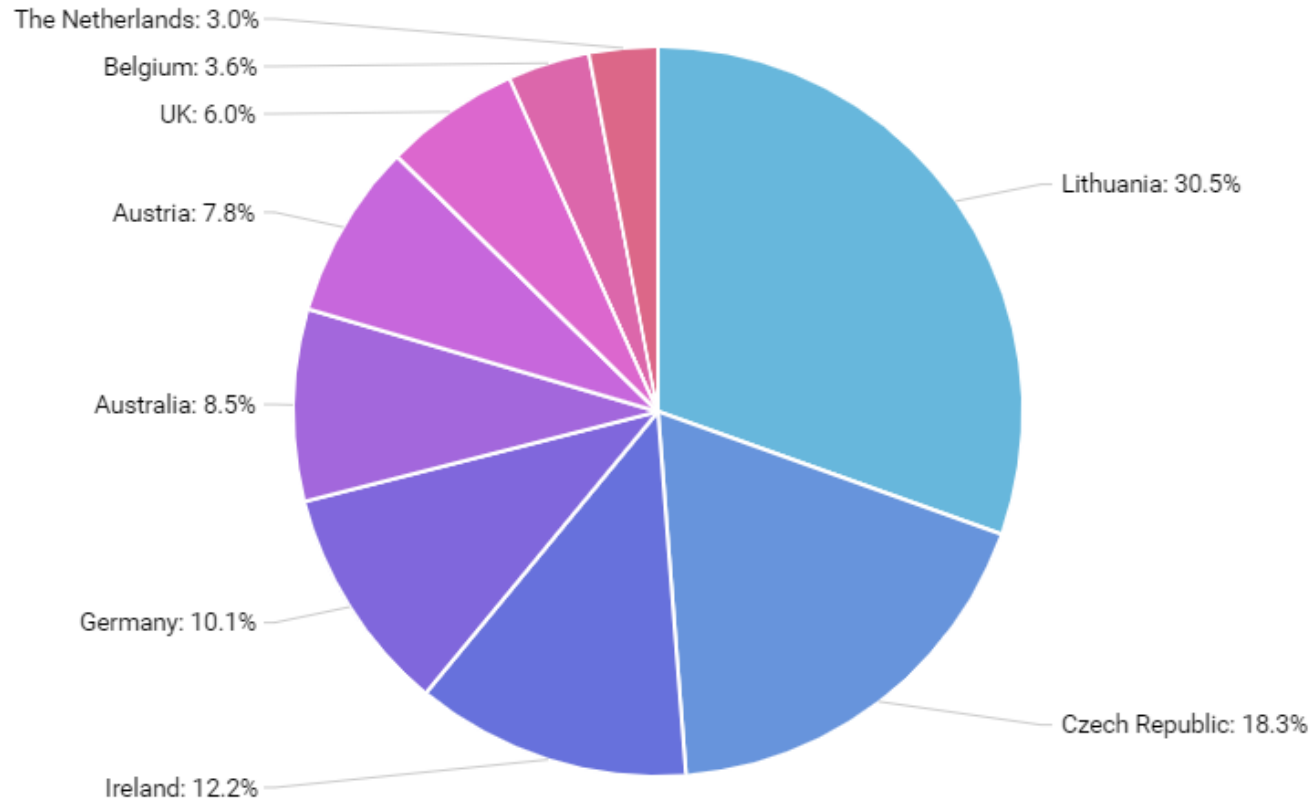


ANGLES ARE HARD TO UNDERSTAND

Avoid pie charts!

Angle judgements are inaccurate

In general, underestimate size of larger angle.



AREAS ARE HARD TO UNDERSTAND

Avoid area charts!

Area judgements are inaccurate

In general, underestimate size of larger area

African Countries by GDP

TOP COUNTRIES BY GDP IN U.S. \$ BILLIONS

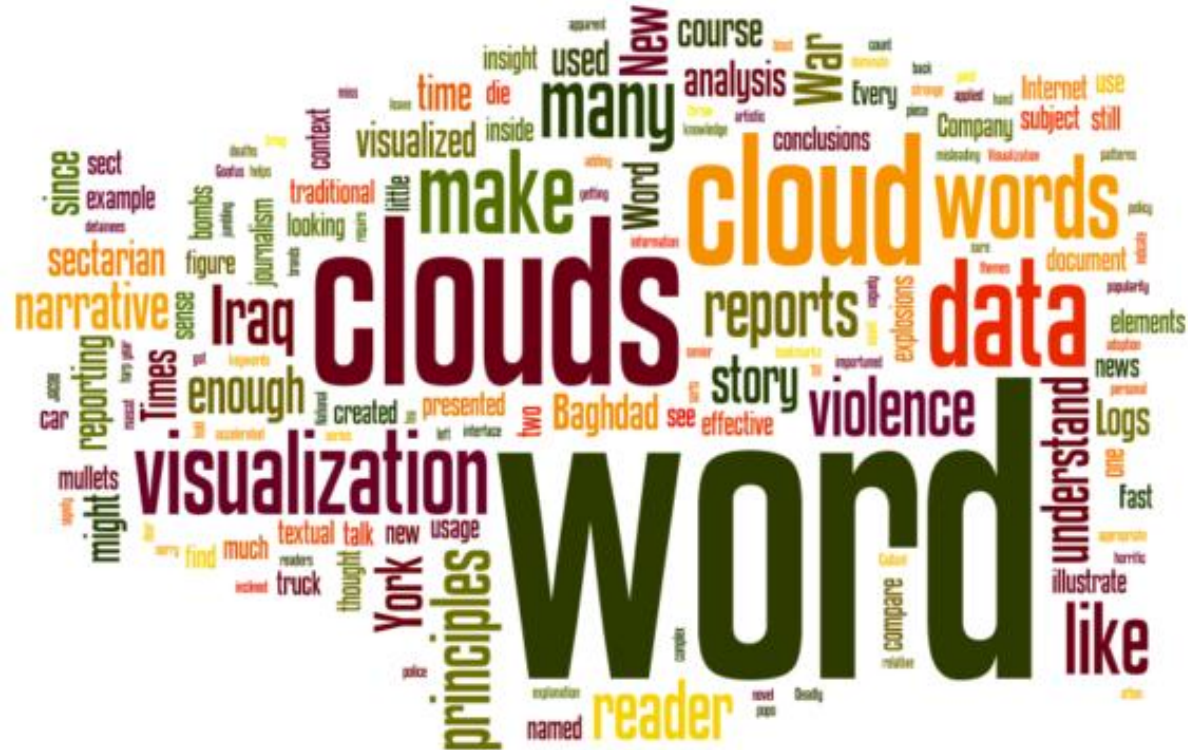
Gross domestic product (GDP) refers to the market value of all final goods and services produced within a country in a given period (2008 - 2009).

GDP CALCULATION

private consumption + gross investment + government spending + exports - imports

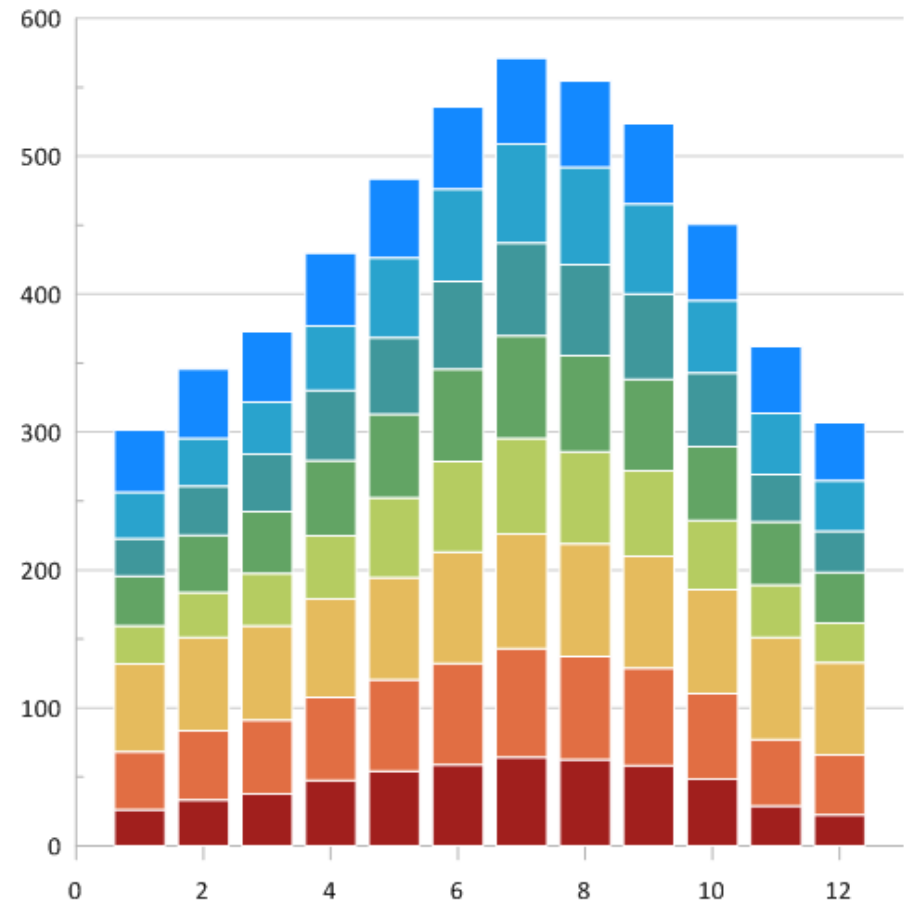


(Here another problem – font size and word length -> area)



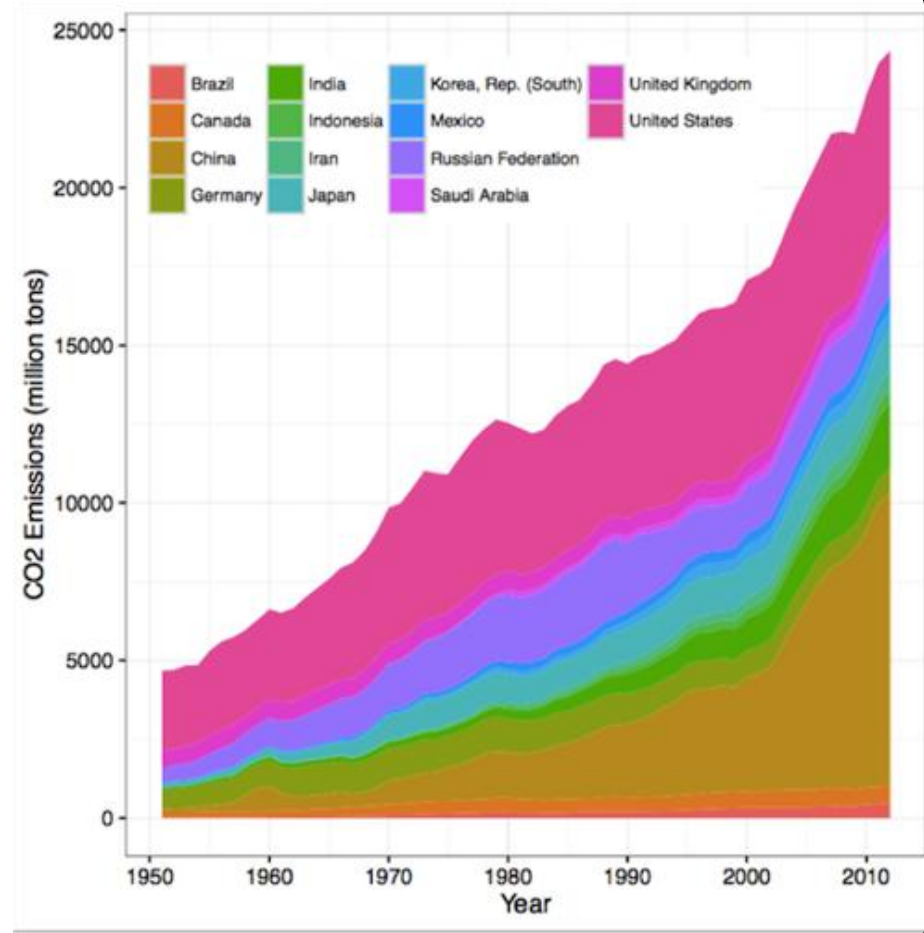
AVOID JIGGLING BASELINE

- Stacked bar charts / histograms hard to read because baseline moves
- Notice that top bars are all about the same height

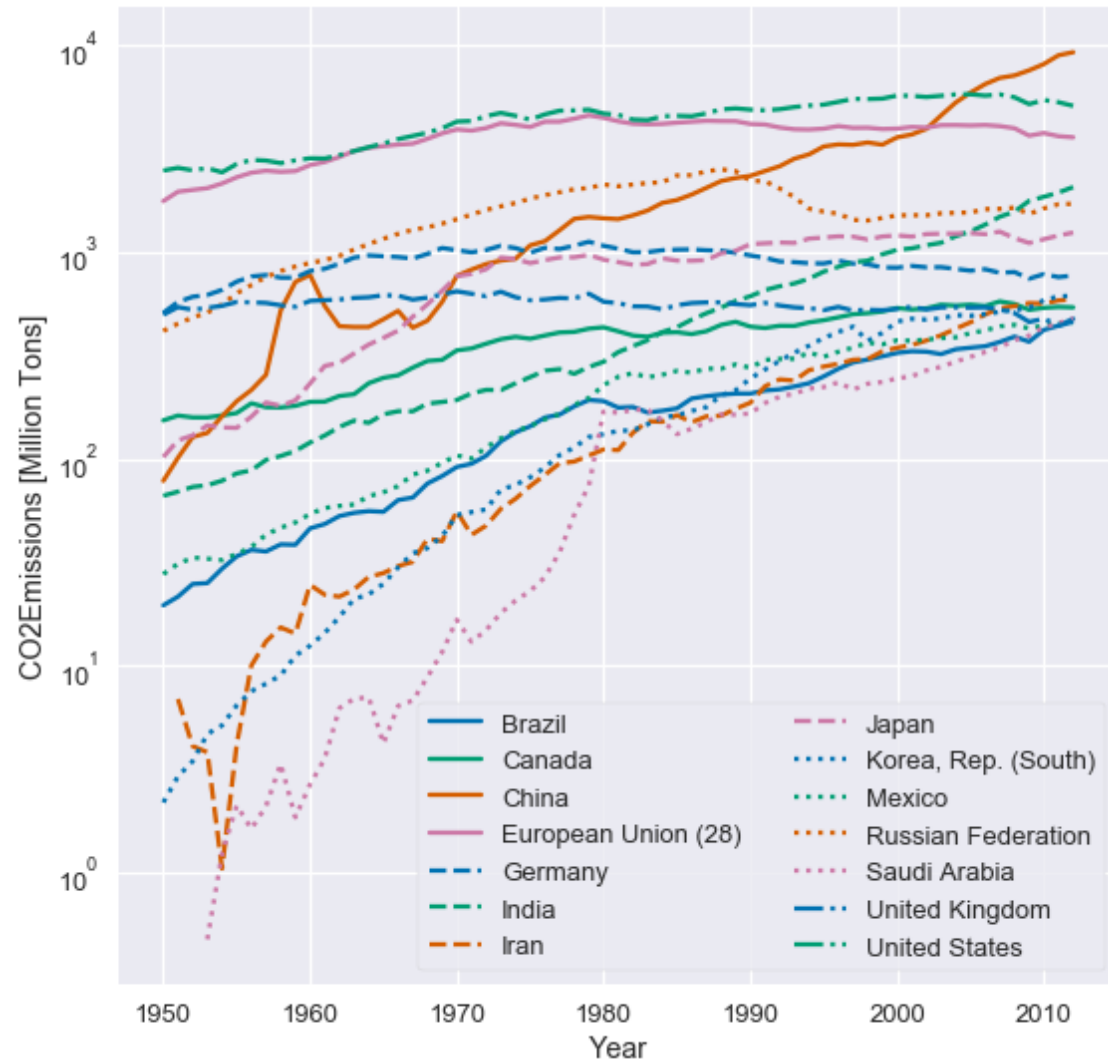


AVOID JIGGLING BASELINE

- **Stacked area charts hard to read because baseline moves**



Instead, plot lines themselves

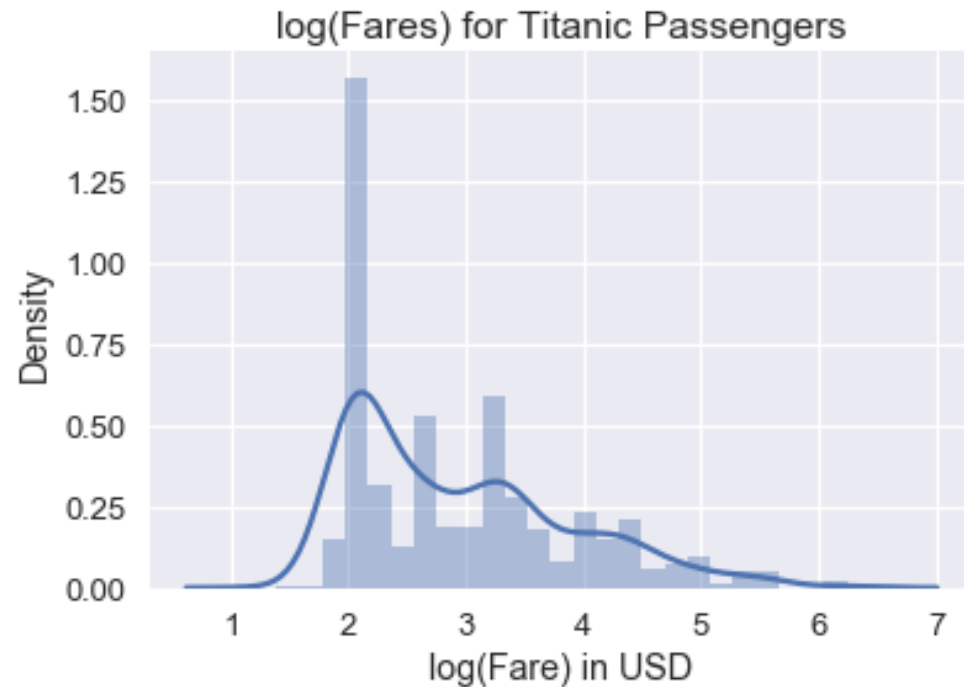
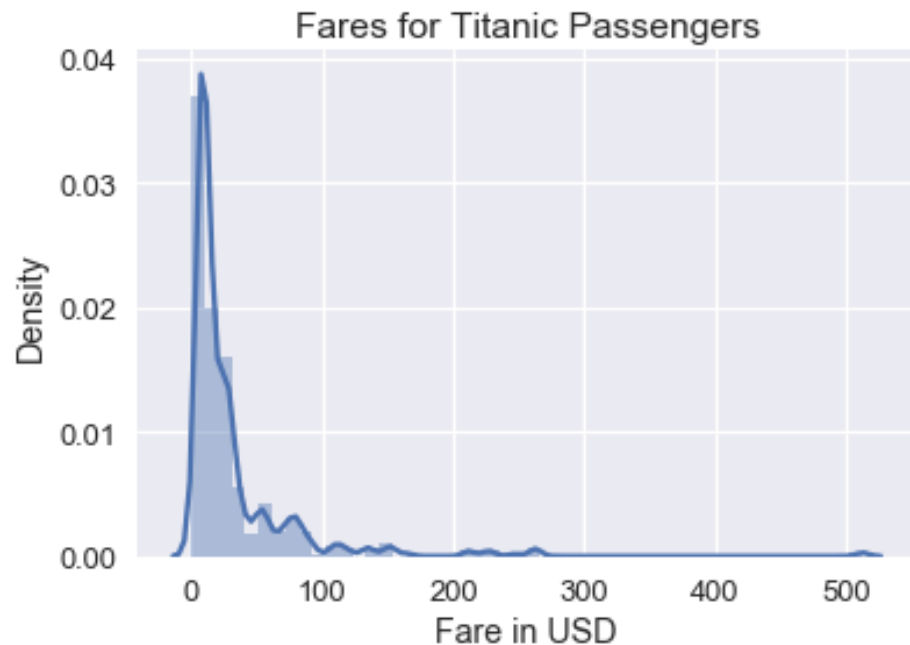


PRINCIPLES OF TRANSFORMATION

Alter data for visualization in order to more
effectively reveal trends

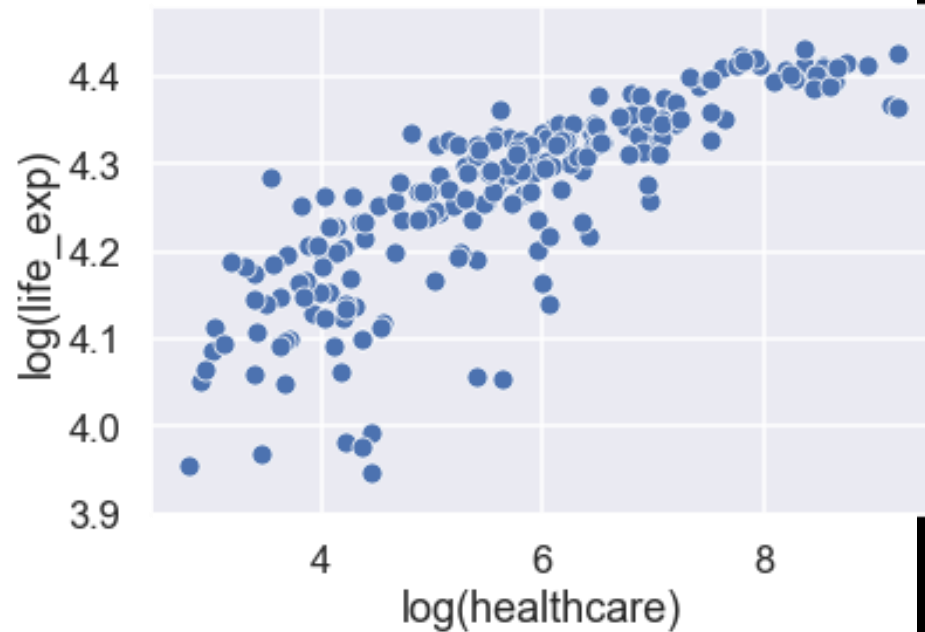
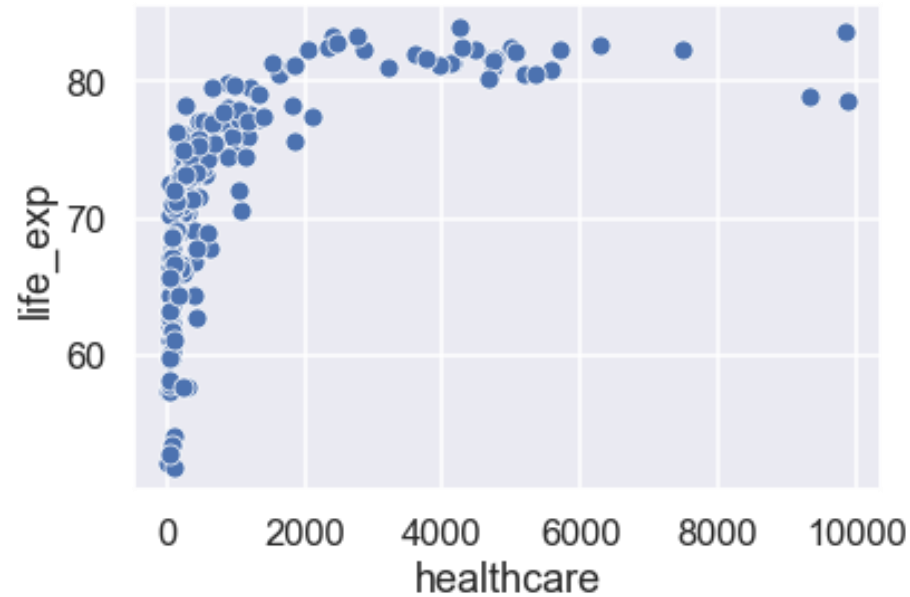
TRANSFORMING DATA CAN REVEAL PATTERNS

- When data are heavy tailed, useful to take the log and replot



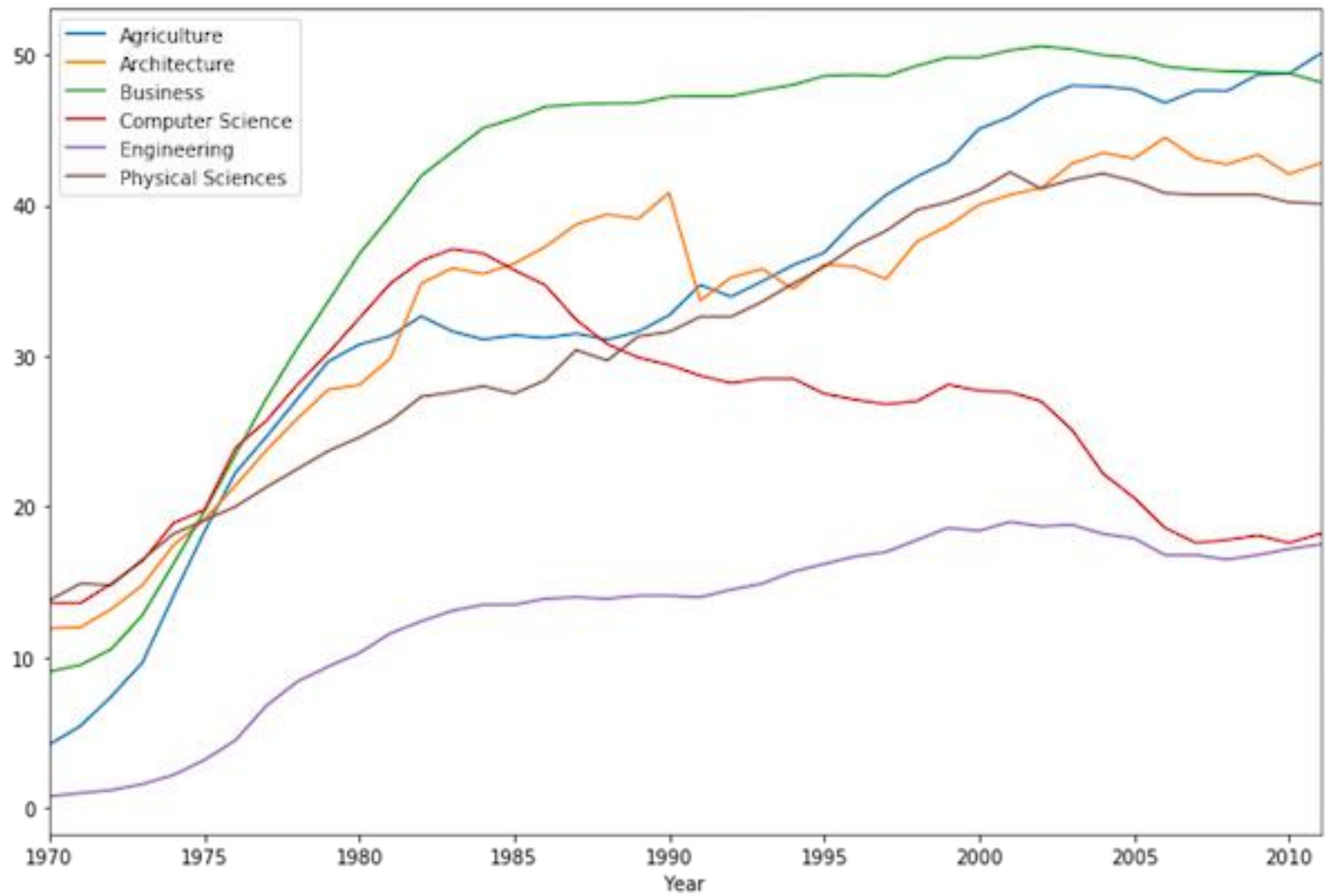
TRANSFORMING DATA CAN REVEAL PATTERNS

- Log of nonlinear data can reveal pattern in scatter plot!



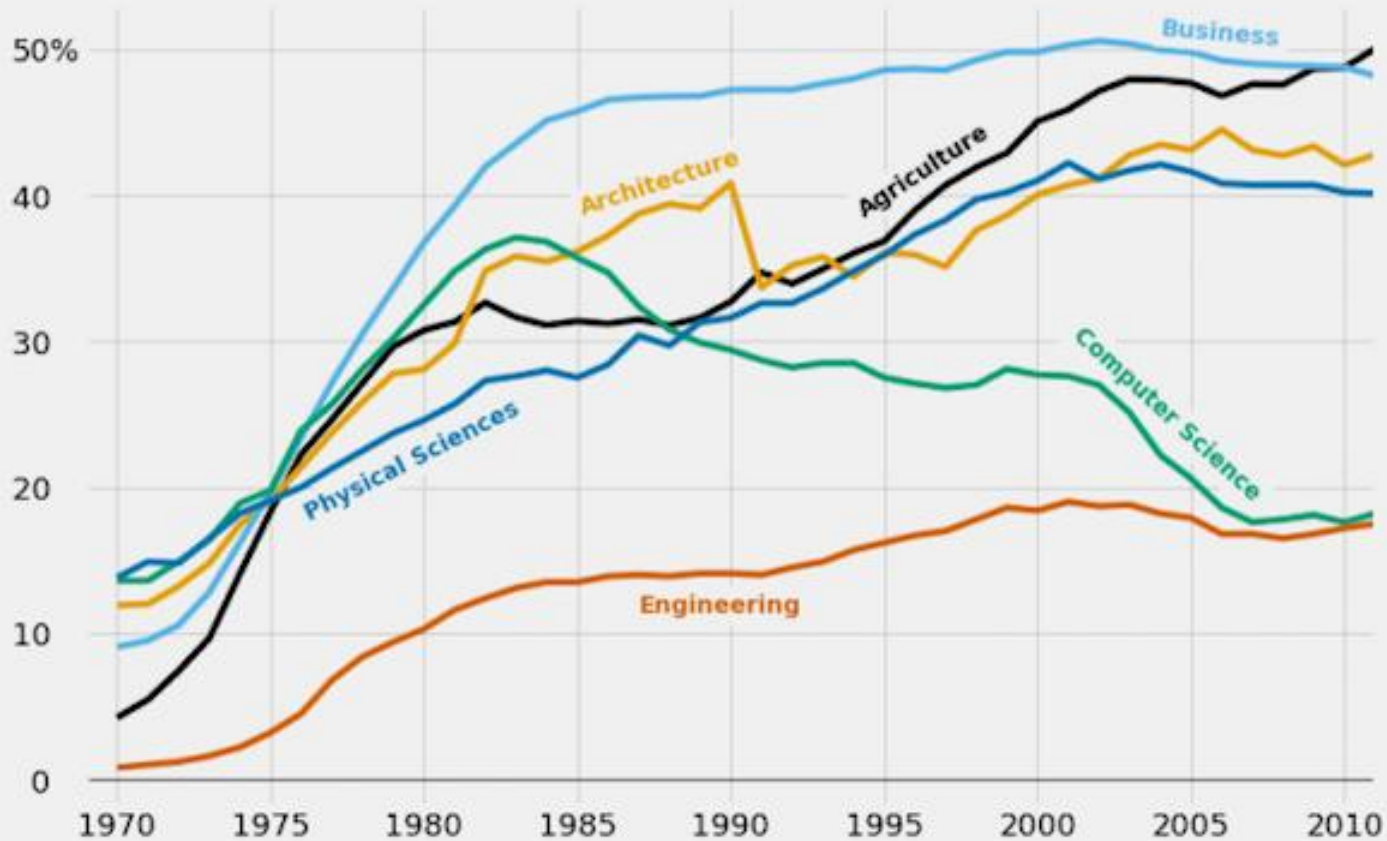
PRINCIPLES OF CONTEXT

Add relevant context to any plot you plan to
share more broadly



The gender gap is transitory - even for extreme cases

Percentage of Bachelors conferred to women from 1970 to 2011 in the US for extreme cases where the percentage was less than 20% in 1970



ADD CONTEXT DIRECTLY TO PLOT

A publication-ready plot needs:

- **Informative title (takeaway, not description)**
 - “Older passengers spend more on plane tickets” instead of “Scatter plot of price vs. age”.
- **Axis labels**
- **Reference lines and markers for important values**
- **Labels for unusual points**
- **Captions that describe data**

PRINCIPLES OF SMOOTHING

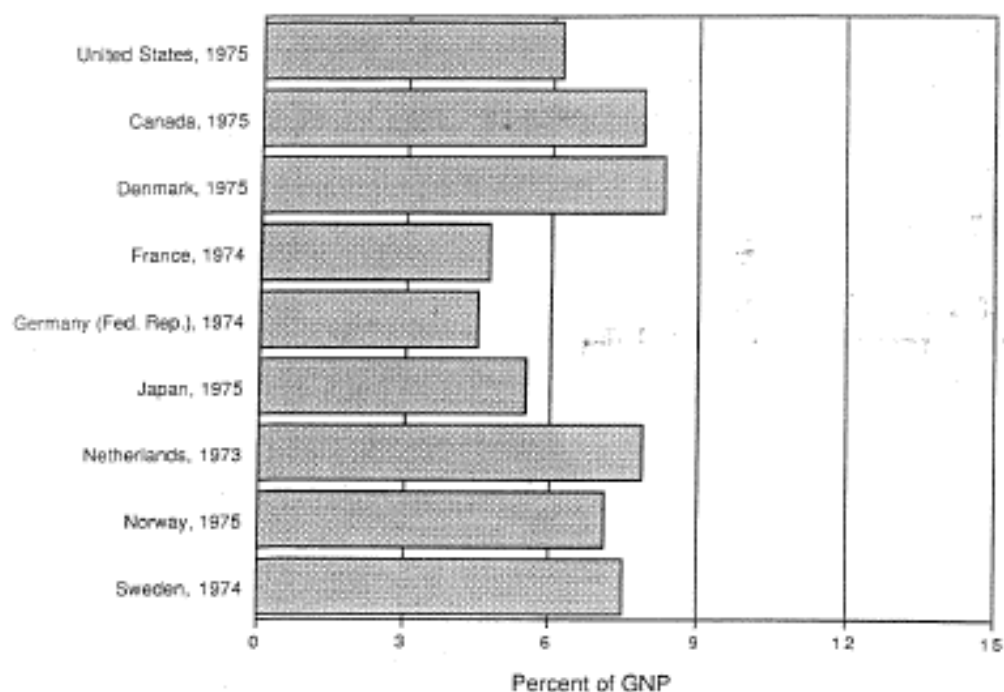
Smoothing allows us to more clearly visualize data
when we have many data points

Howard Wainer: How to Display Data Badly

**THE AIM OF
GOOD DATA GRAPHICS
IS TO
DISPLAY DATA
ACCURATELY AND CLEARLY.**

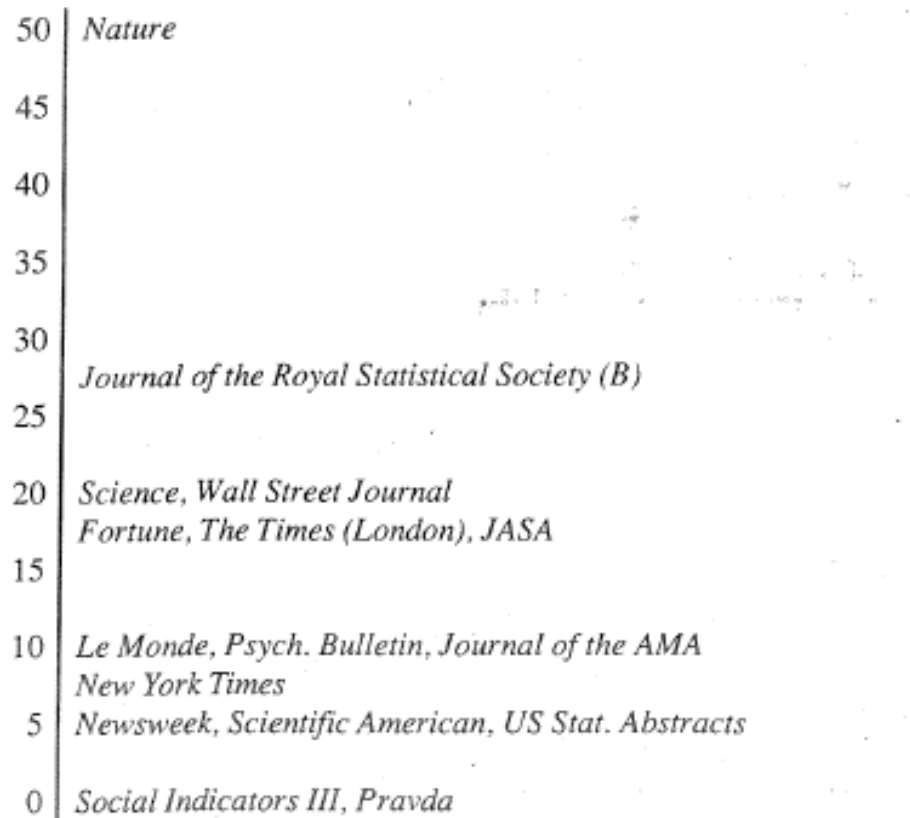
**I. INCLUDE AS FEW DATA AS
POSSIBLE
(MINIMIZE THE DATA
DENSITY)**

Chart 6/25. Expenditures for Education as a Percent of GNP, Selected Countries: Mid-1970's



Data Density = 9 numbers / 63 sq. ins. = .14

Median Data Densities for the Graphics in Some Well-Known Publications



Source: Tufte, 1983

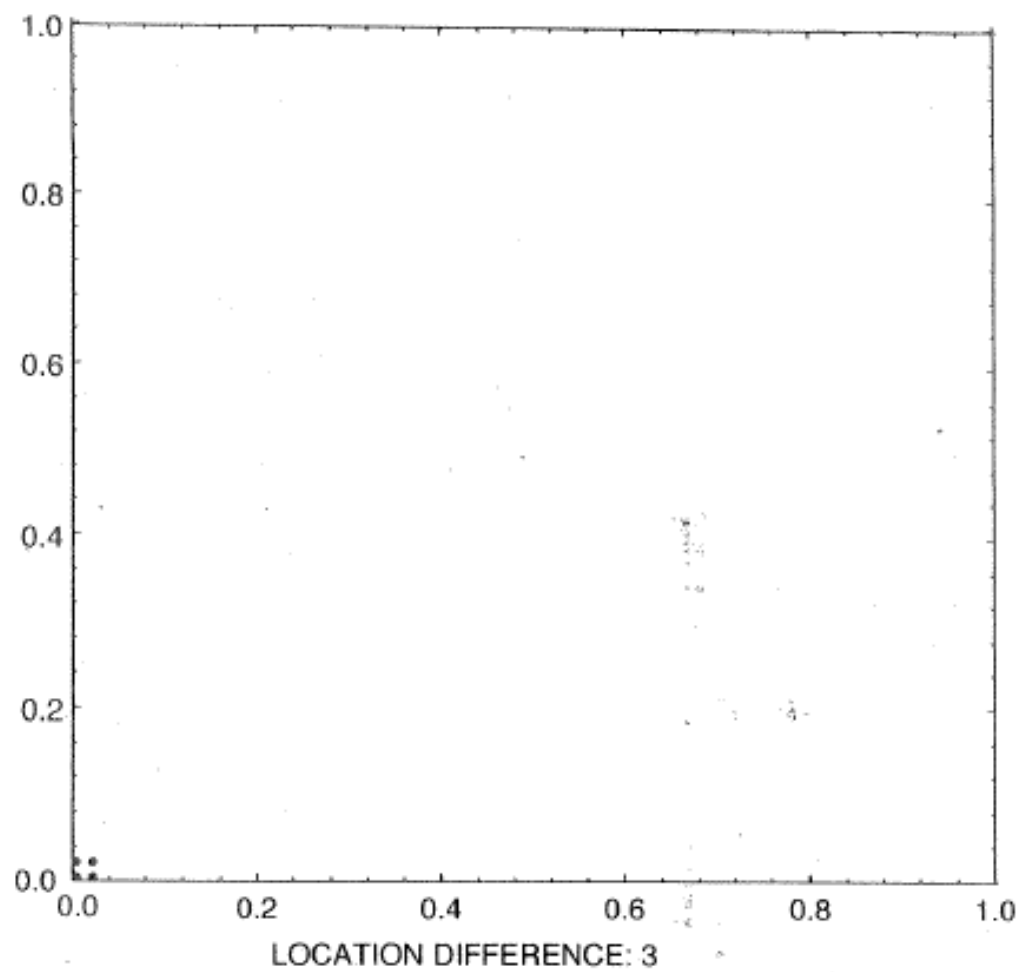
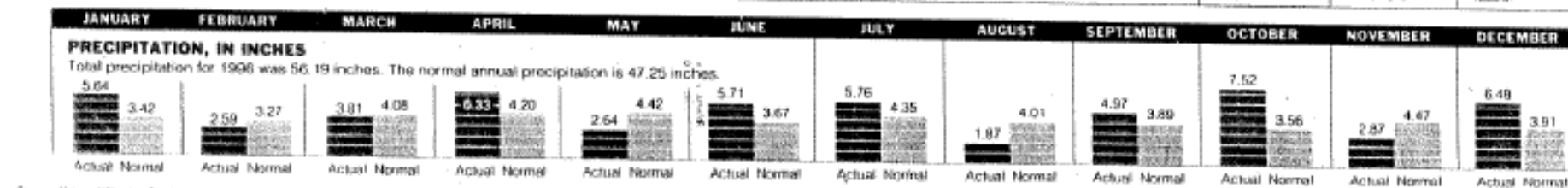
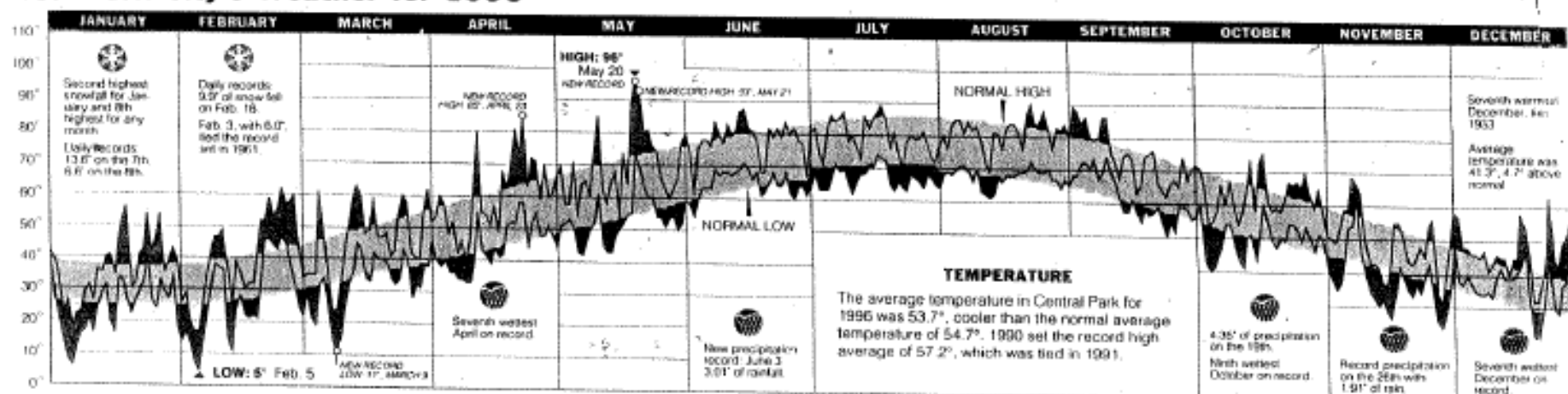


Figure 1. Coordinate Partial Order P-P Plot

New York City's Weather for 1996



Source: National Weather Service

CHART JUNK

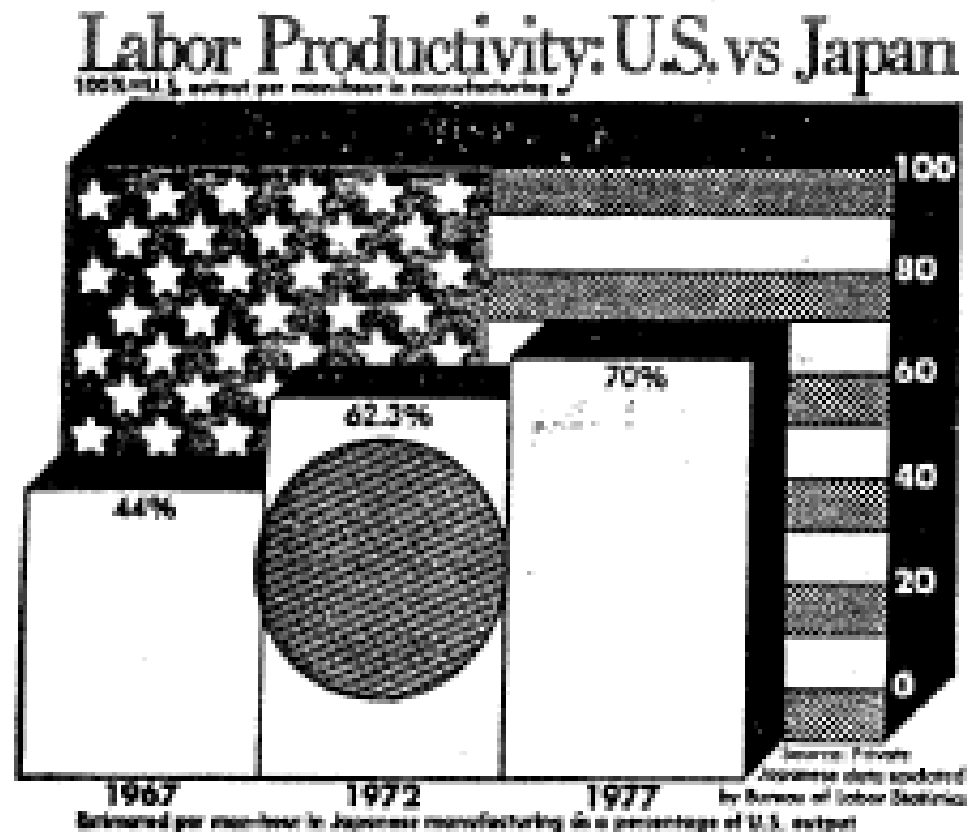
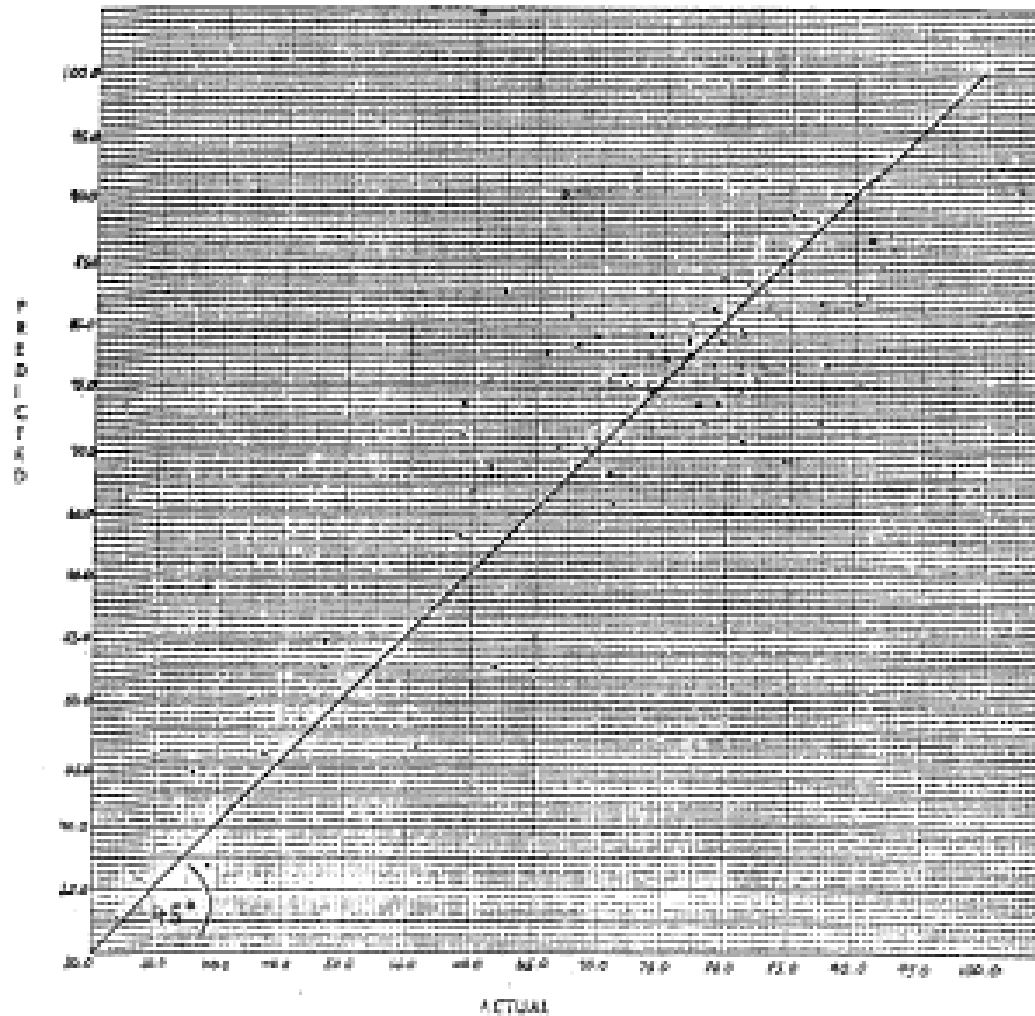


Figure 3. A low density graph (© 1978, The Washington Post) with chart-junk to fill in the space ($ddl = .2$).

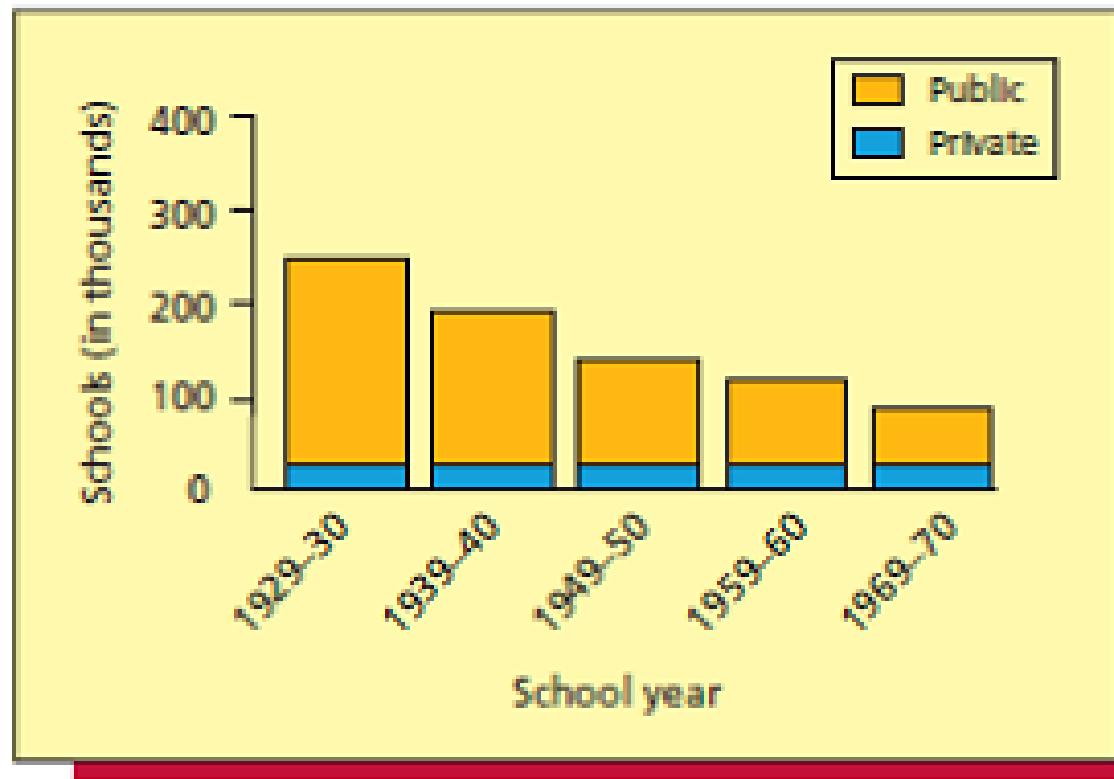
II. HIDE WHAT DATA YOU DO SHOW (MINIMIZE THE DATA-INK RATIO)

HIDE THE DATA IN THE GRID

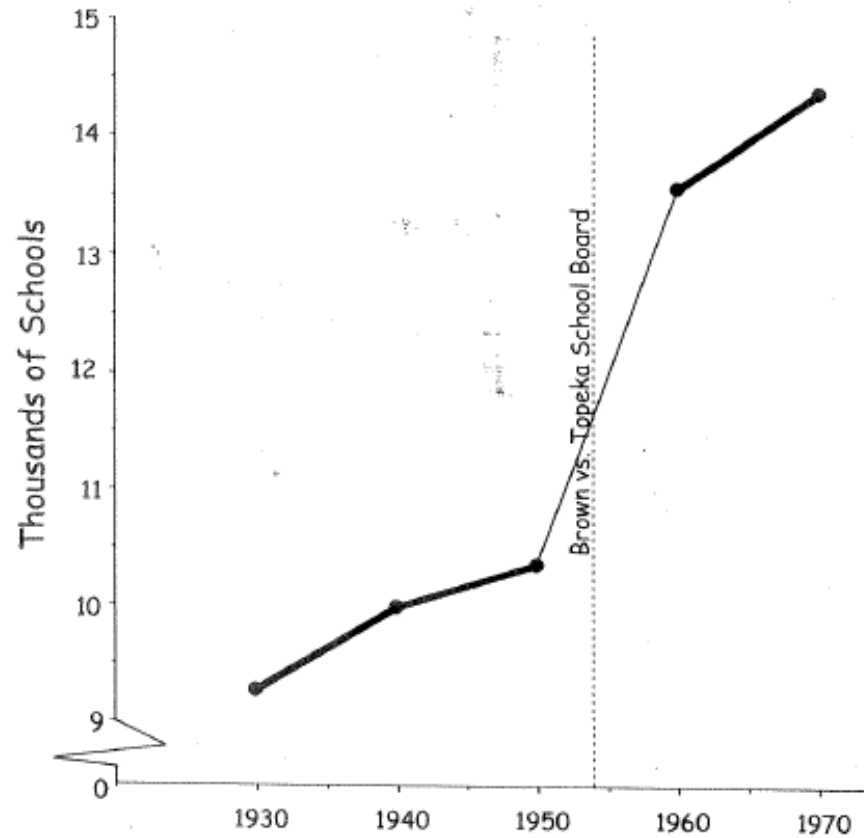
Relationship of Actual Rates of Registration to Predicted Rates
(104 cities 1960).



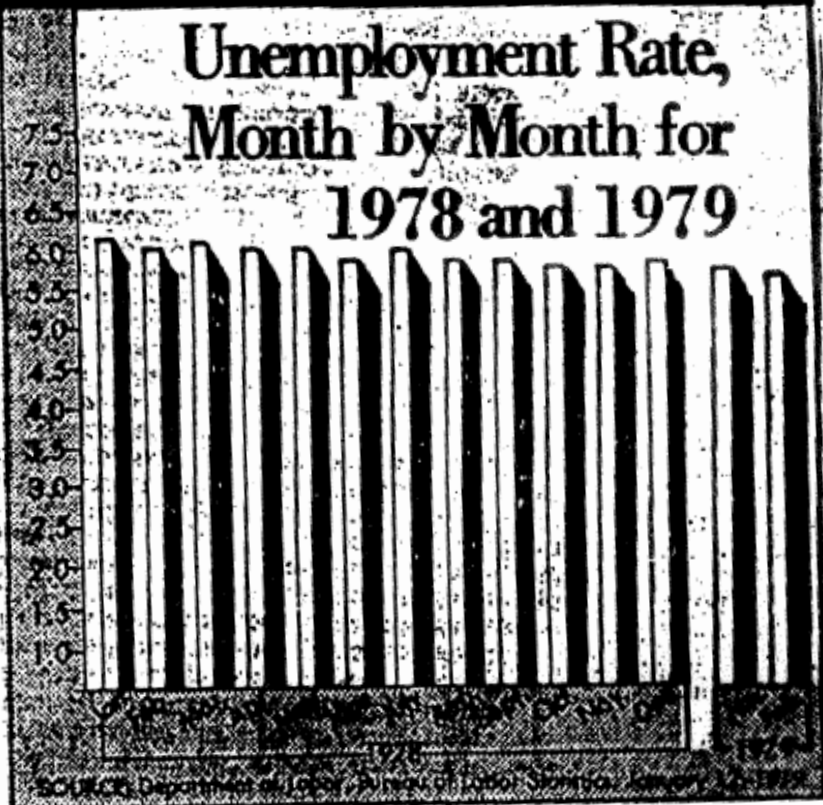
HIDE THE DATA IN THE SCALE



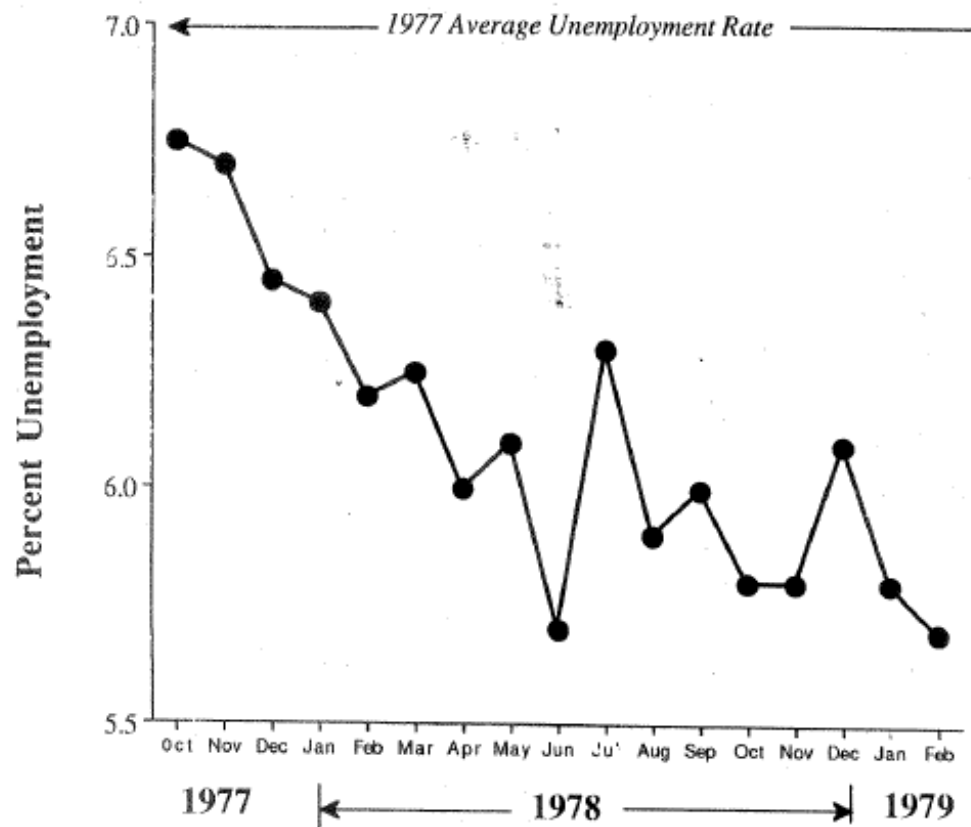
Number of Private Elementary Schools in the
United States from 1930 until 1970



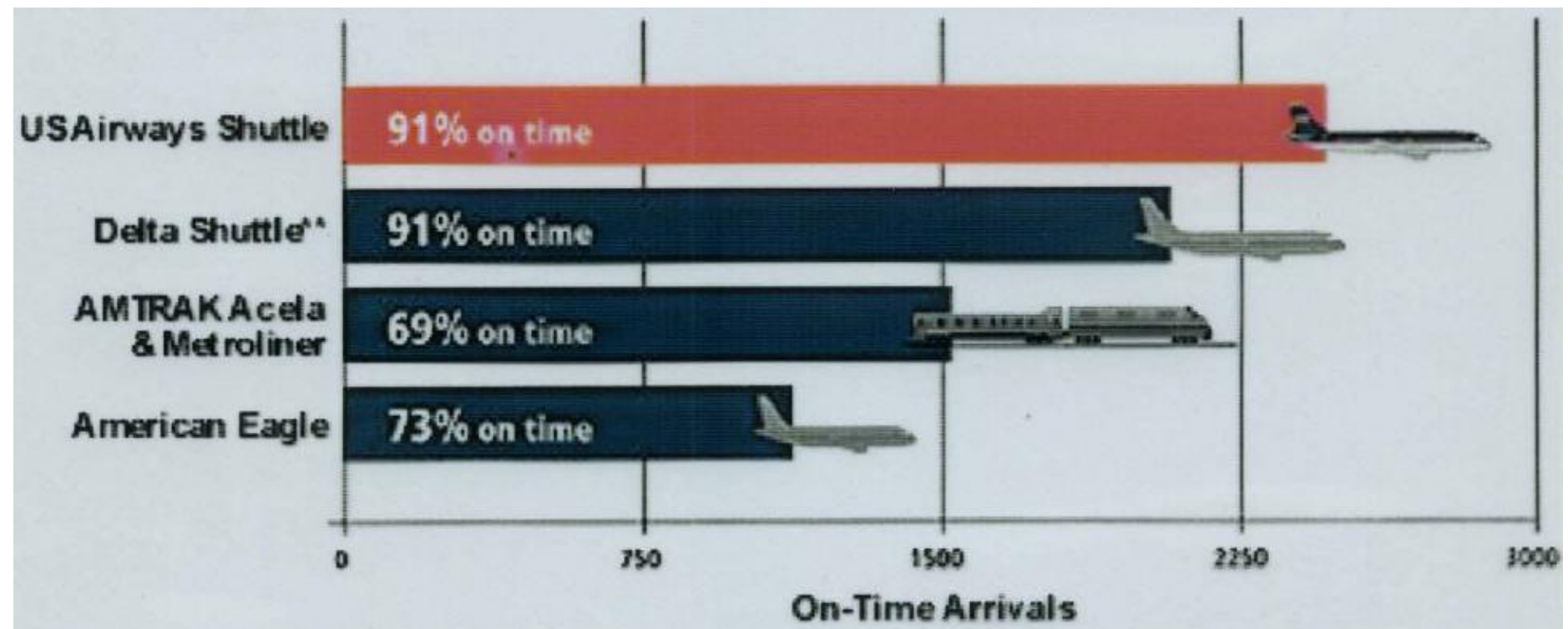
Unemployment Rate, Month by Month for 1978 and 1979



Unemployment Rate by Month Since 1977

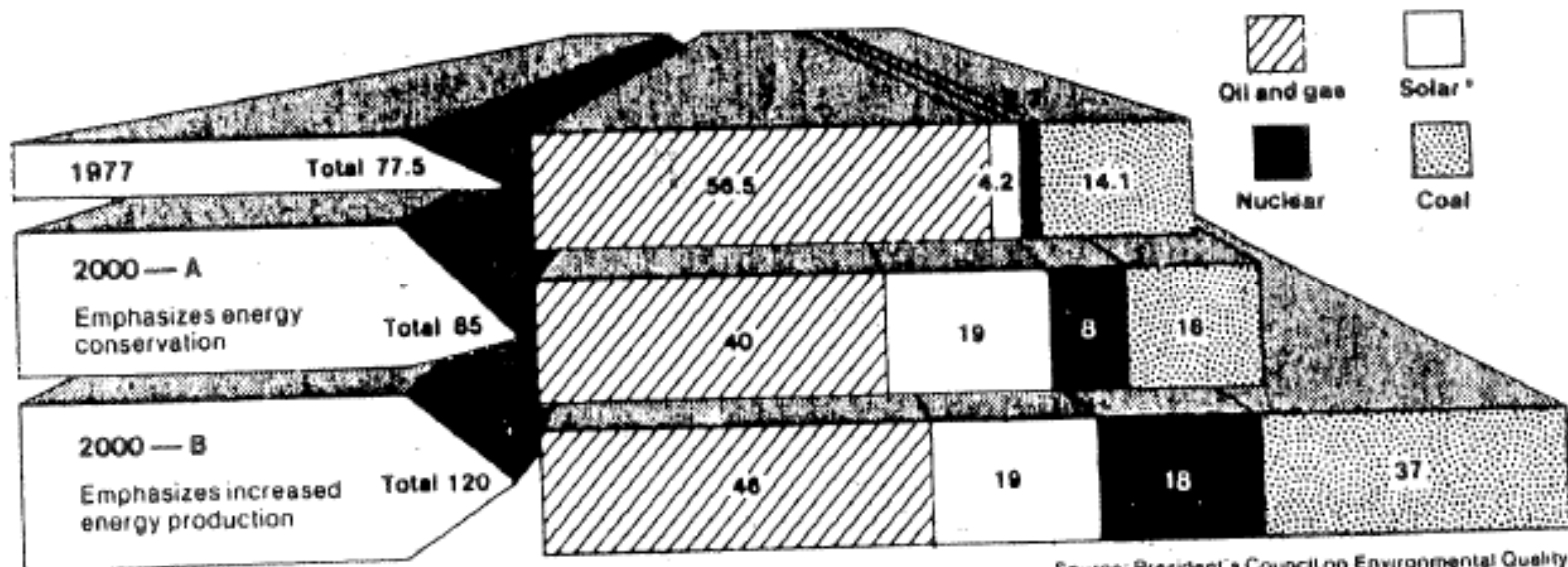


III. IGNORE THE VISUAL METAPHOR



A New Set of Projections for the U.S. Supply of Energy

Compared are two projections of United States energy supply in the year 2000 made by the President's Council of Environmental Quality and the actual 1977 supply. All figures are in "quads," units of measurement that represent a million billion — one quadrillion — British thermal units (B.T.U.'s), a standard measure of energy.



* Includes all renewable energy sources

Source: President's Council on Environmental Quality

הצעירים בישראל בחרו - מהפך!

תוצאות הצבעת תלמידי י"ב **maiko**

30%

המחנה
הציוני

20%

יש עתיד

9%

הליכוד

8%

כולנו

7%

מרצ

6%

הבית היהודי



המחנה הציוני בראשות הרצוג | לבני

Like

Comment



יצחק (בוז'י) הרצוג -
יש עתיד.

Like · Comment · Share · 26 February

Album: Timeline Photos

Shared with: Public

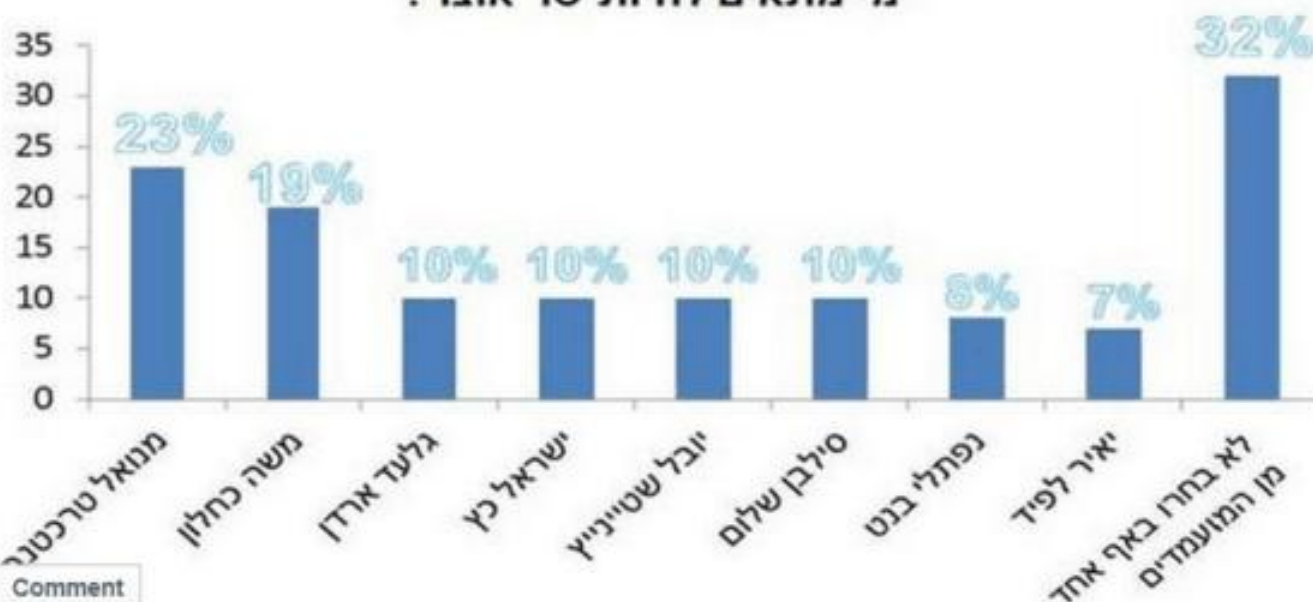
סקר גלצ: 23% רוצים את טרכטנברג במשרד האוצר

מי הכי מתאים להיות שר האוצר? על פי סקר שנערך עבור גלצ עולה כי טרכטנברג ונחלון בראש הרשימה, עם 23% ו-19% בהתאמה. אחריהם השרים שטייניץ, ארדן, כץ ושלוש עם 10% כל אחד. בנט ולפיד הרחק מאחור

עידו בנבג'י



מי מתאים להיות שר אוצר?



Like Comment



המחנה הציוני

הציבור התפכח ומבין שמשרד האוצר צריך איש מקצוע - ולא פרונטור. את כל המסקנות מהוועדה שקמה לאחר המחאה החברתית בראשותו, מהן נתניהו התעלם, רמס וגרס - אנחנו מתכוונים ליישם.

פרופ' מנואל טרכטנברג - Prof. Manuel Trajtenberg - שר האוצר שלנו - בדרך למהפך הכלכלה הישראלית

From: המחנה הציוני's photos in Timeline Photos

Shared with: Public

Open Photo Viewer

Download

U.S. trade with China and Taiwan

6-14-81

(in millions of U.S. dollars)

3,000

U.S. exports
to China

2,000

U.S. imports
from China

1,000

1972 1974 1976 1978 1980

(in millions of U.S. dollars)

6,000

U.S. imports
from Taiwan

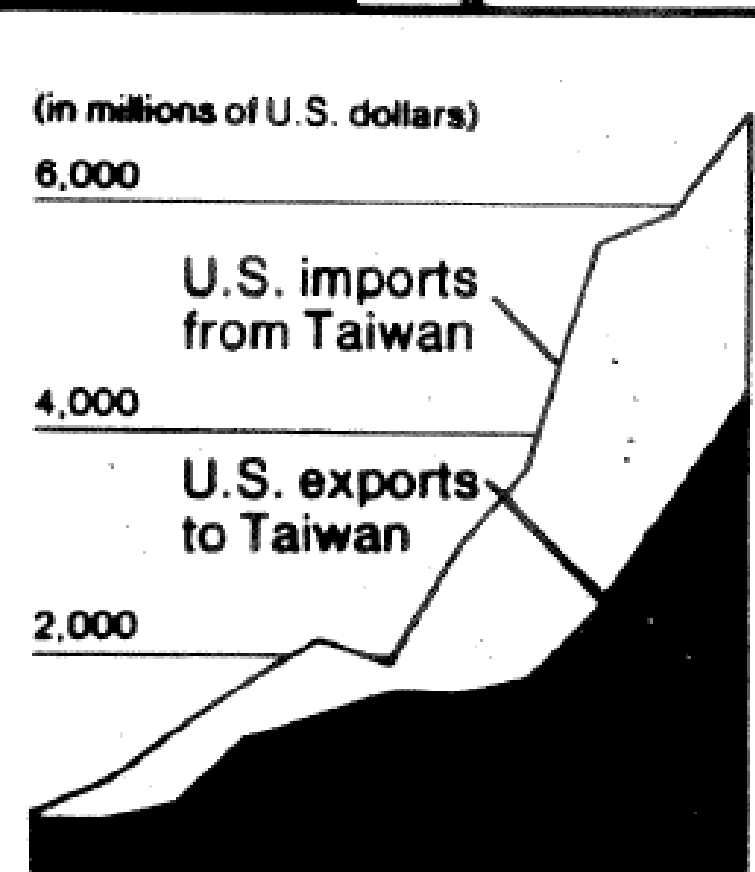
4,000

U.S. exports
to Taiwan

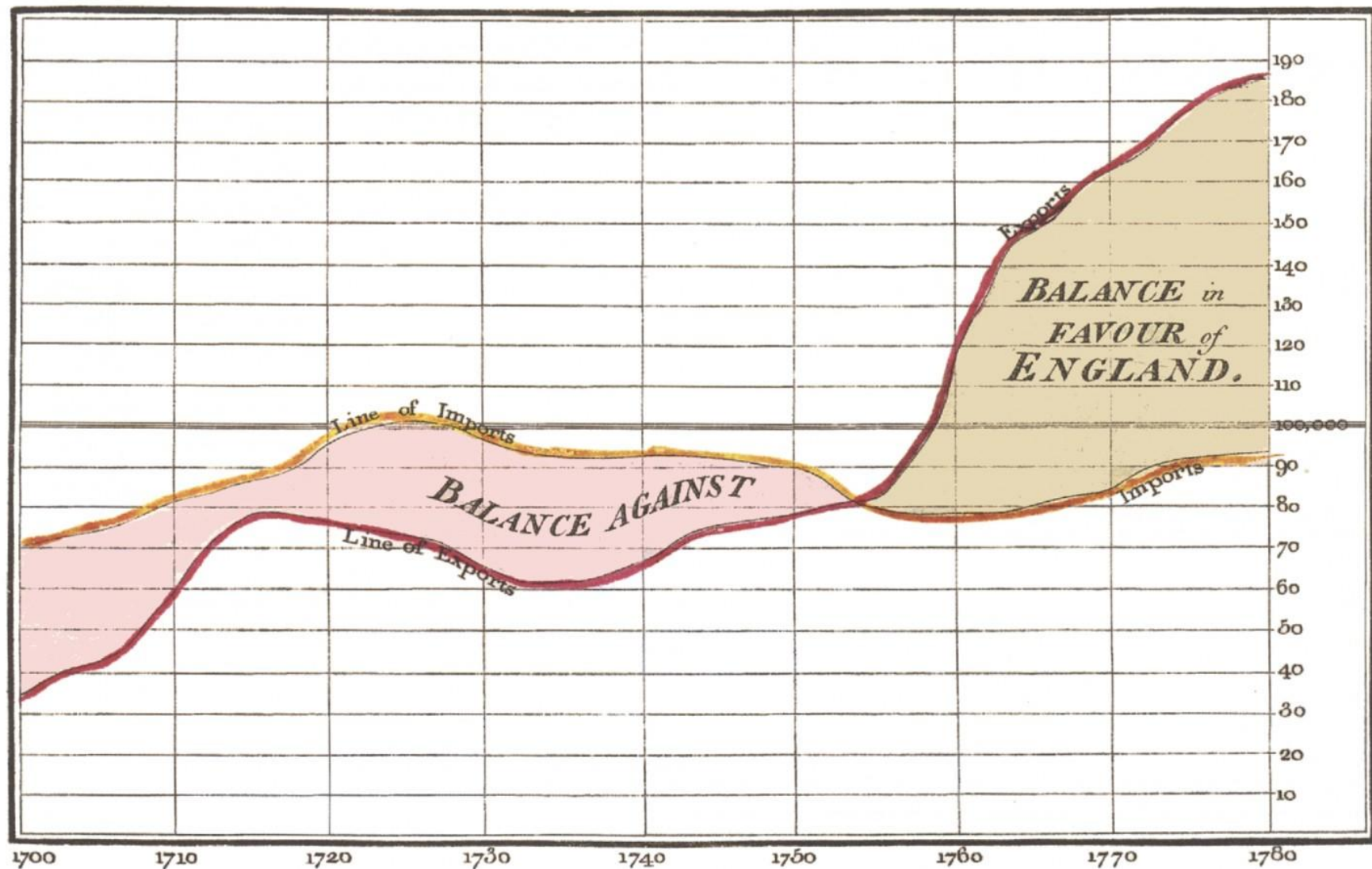
2,000

1970 1972 1974 1976 1978 1980

Source: Department of Commerce



Exports and Imports to and from DENMARK & NORWAY from 1700 to 1780.



The Bottom line is divided into Years, the Right hand line into £10,000 each.

Published as the Act directs, 14th May 1786. by W^m Playfair

Neele sculpt 352, Strand, London.

SHOW THE DATA ACCURATELY



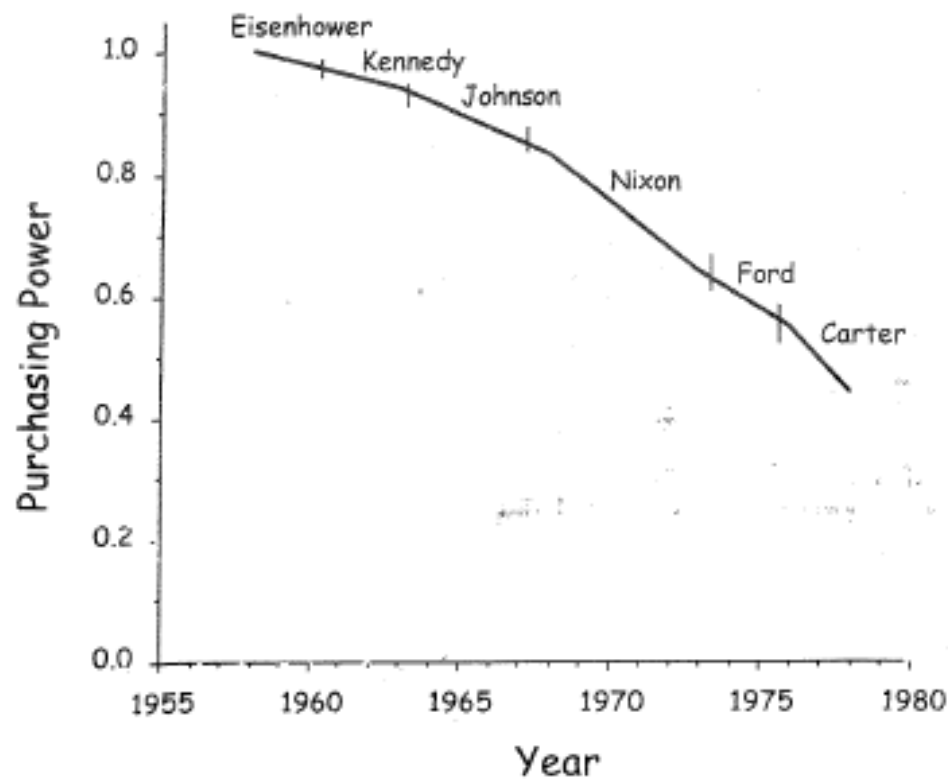
Goosing up the effect By
Squaring the Eyeball
Trick

$$\begin{array}{l} \text{Actual} \\ \frac{1.00 - 0.44}{0.44} = 1.27 \end{array}$$

$$\begin{array}{l} \text{Measured} \\ \frac{22.00 - 2.06}{2.06} = 9.68 \end{array}$$

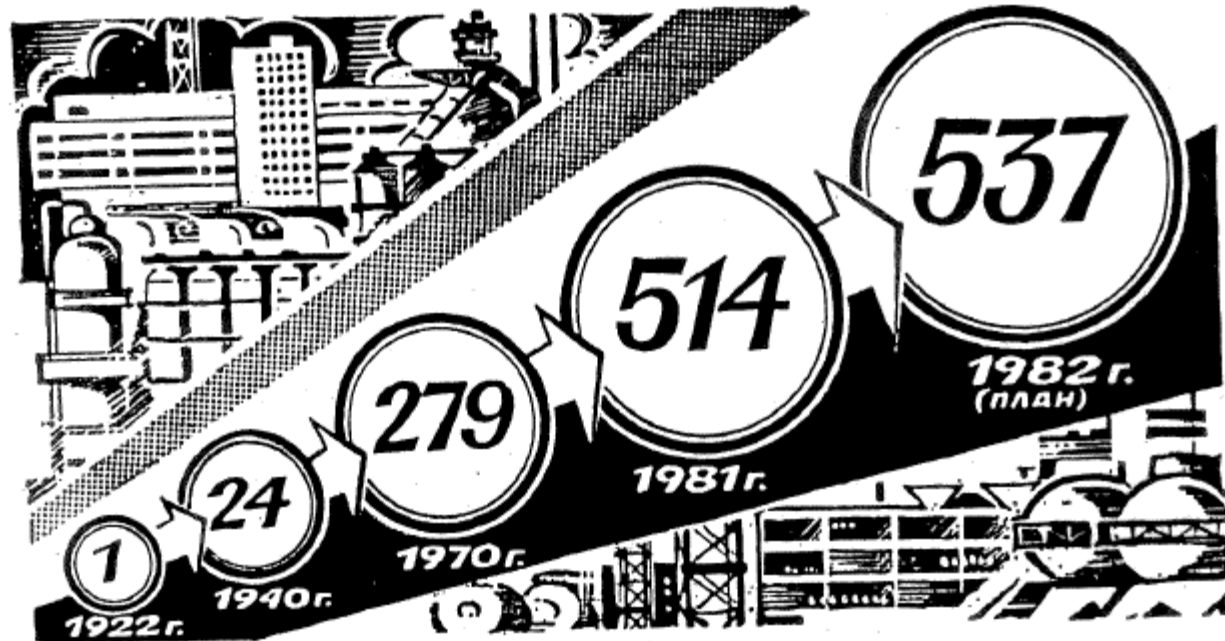
$$\text{Lie Factor} = 9.68 / 1.27 = 762\%$$

The purchasing power of the dollar declined most sharply during the Nixon administration



IV. ONLY ORDER MATTERS (MAXIMIZE THE LIE FACTOR)

THE PRAVDA SCHOOL OF ORDINAL GRAPHICS

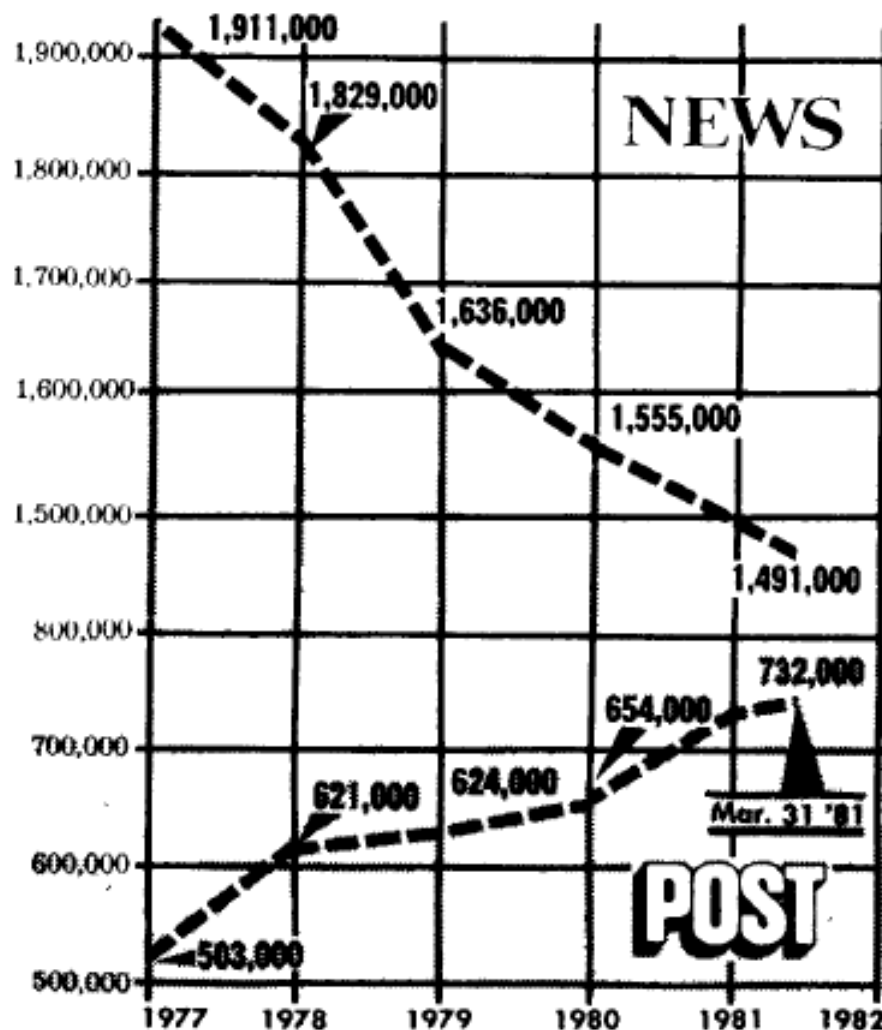


Pravda, May 24, 1982.

SHOW THE DATA CLEARLY

VI. CHANGE SCALES IN MID-AXIS

The soaraway Post — the daily paper New Yorkers trust

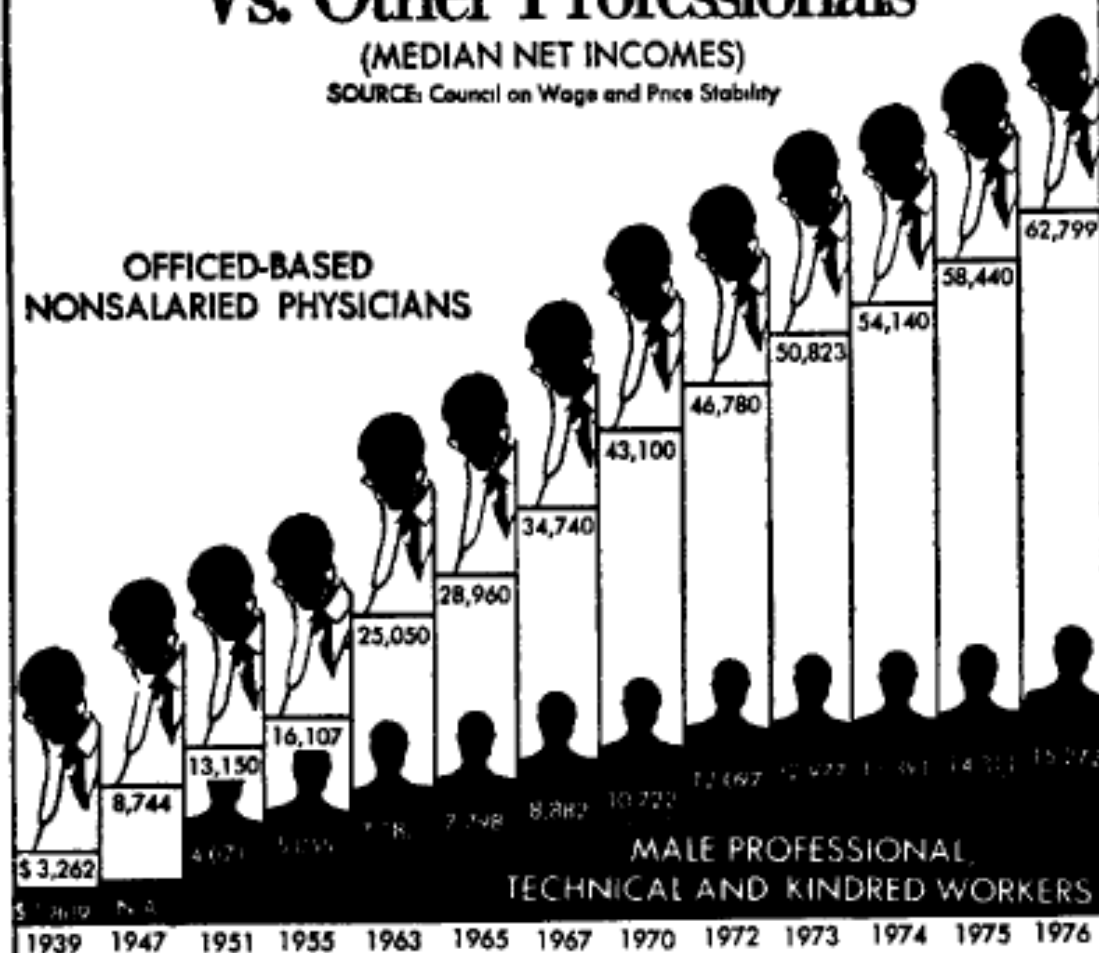


Incomes of Doctors Vs. Other Professionals

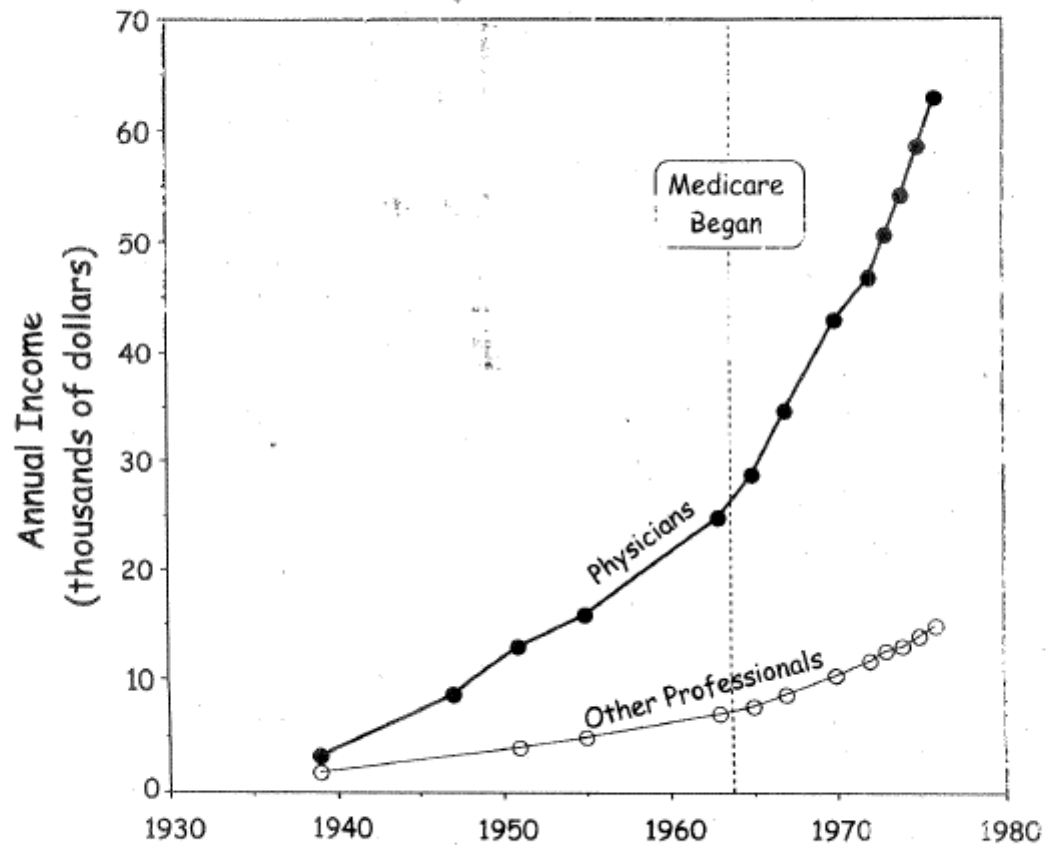
(MEDIAN NET INCOMES)

SOURCE: Council on Wage and Price Stability

OFFICED-BASED
NONSALARIED PHYSICIANS

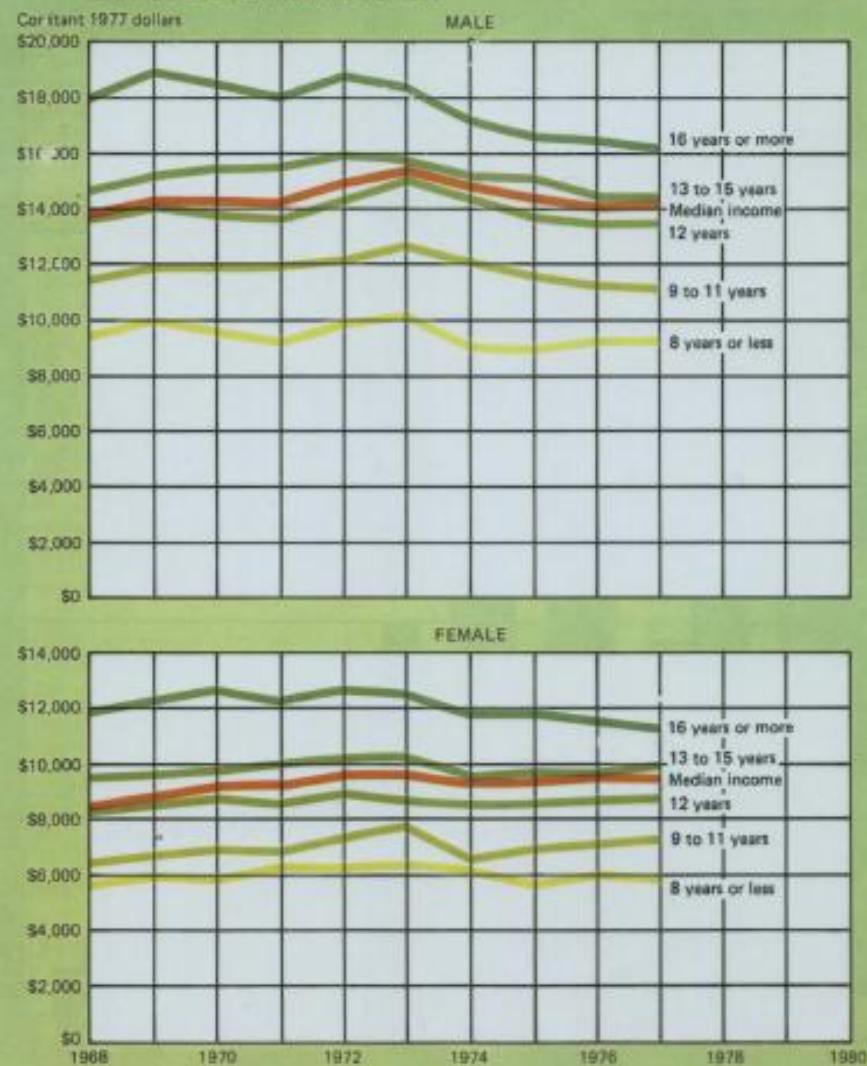


Physicians' Incomes Have Increased at More Than the Rate of Those of Other Professionals



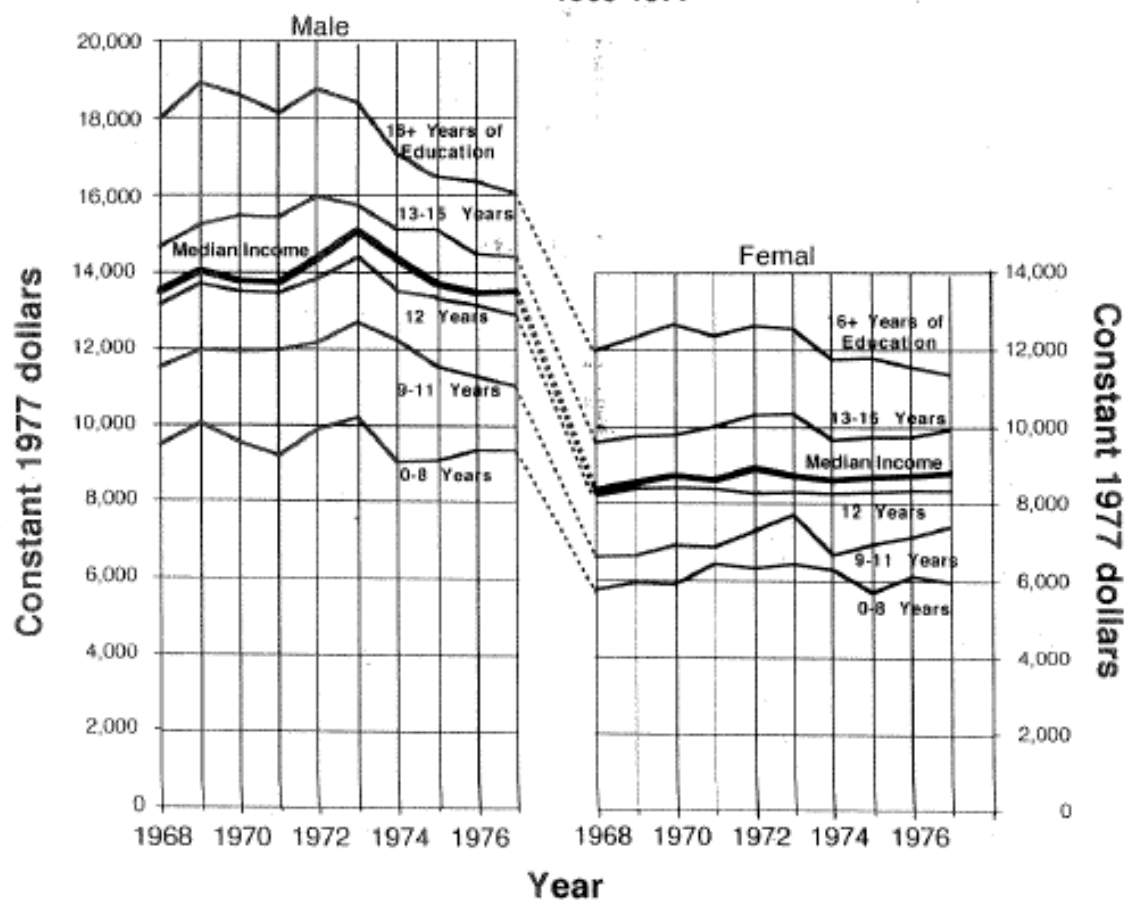
VII. EMPHASIZE THE TRIVIAL (IGNORE THE IMPORTANT)

Chart 6/19. **Median Income of Year-Round, Full-Time Workers 25 to 34 Years Old, by Sex and Educational Attainment: 1968-1977**



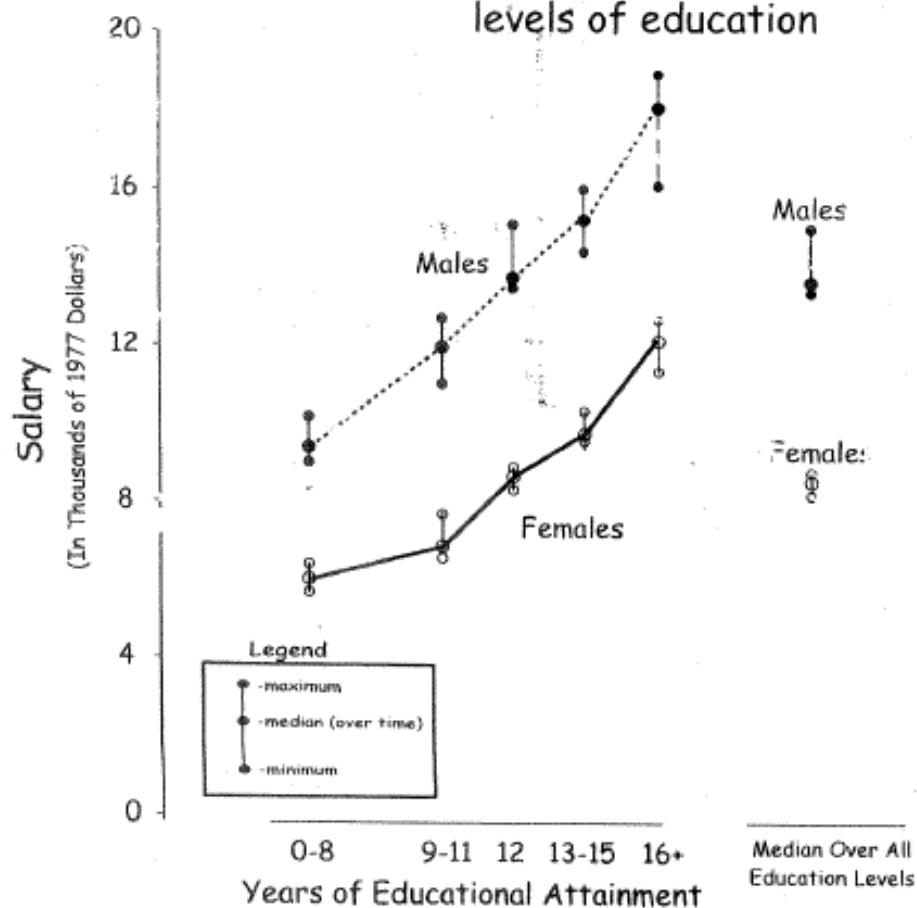
Source: See table 6/19.

**Median Income of Year-Round Full Time Workers
25-34 Years Old by Sex and Educational Attainment:
1968-1977**



Source: *Social Indicators III*
Table 6/19

In the period 1968-1977 women's salaries were about 60% of men's at comparable levels of education

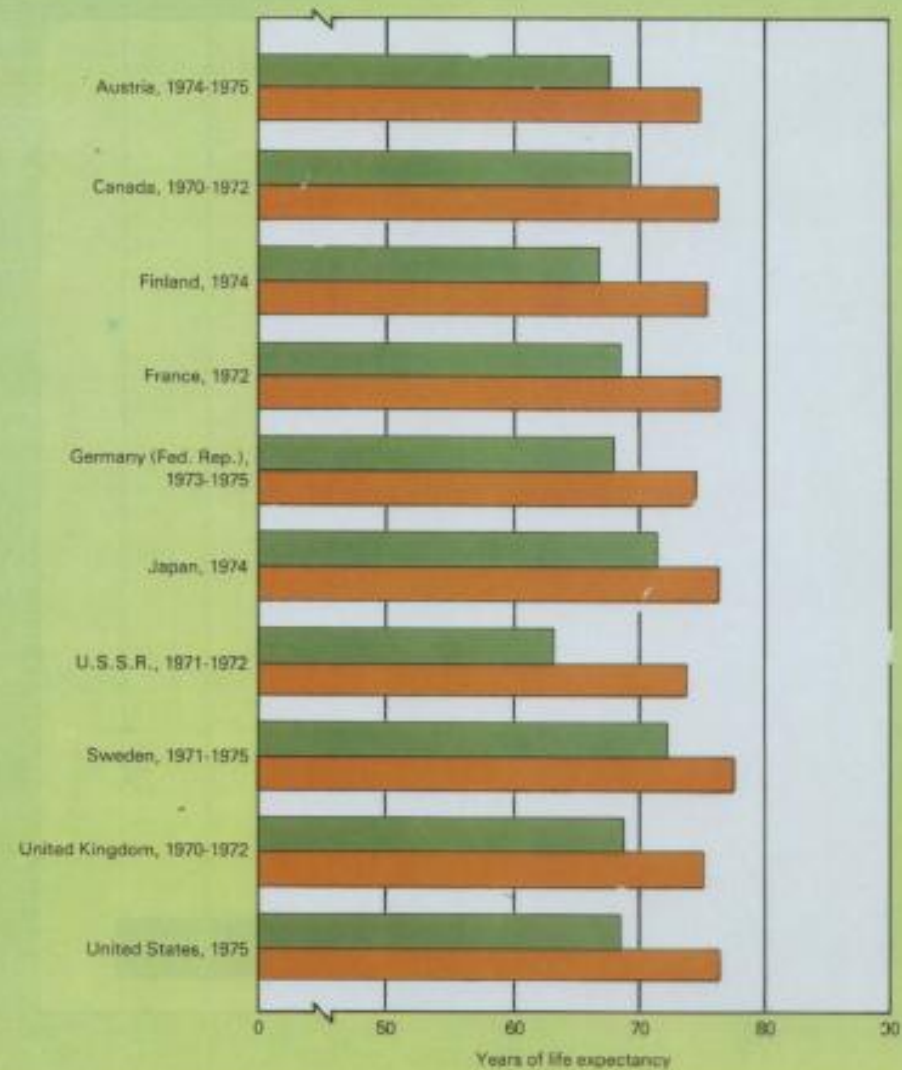


Source: Social Indicators III
Table 6/19

IX. AUSTRIA FIRST

Chart 2/29. **Life Expectancy at Birth, by Sex, Selected Countries, Most Recent Available Year: 1970-1975**

Male
Female



Source: See table 2/29.

Life Expectancy at Birth, By Sex Most Recent Available Year

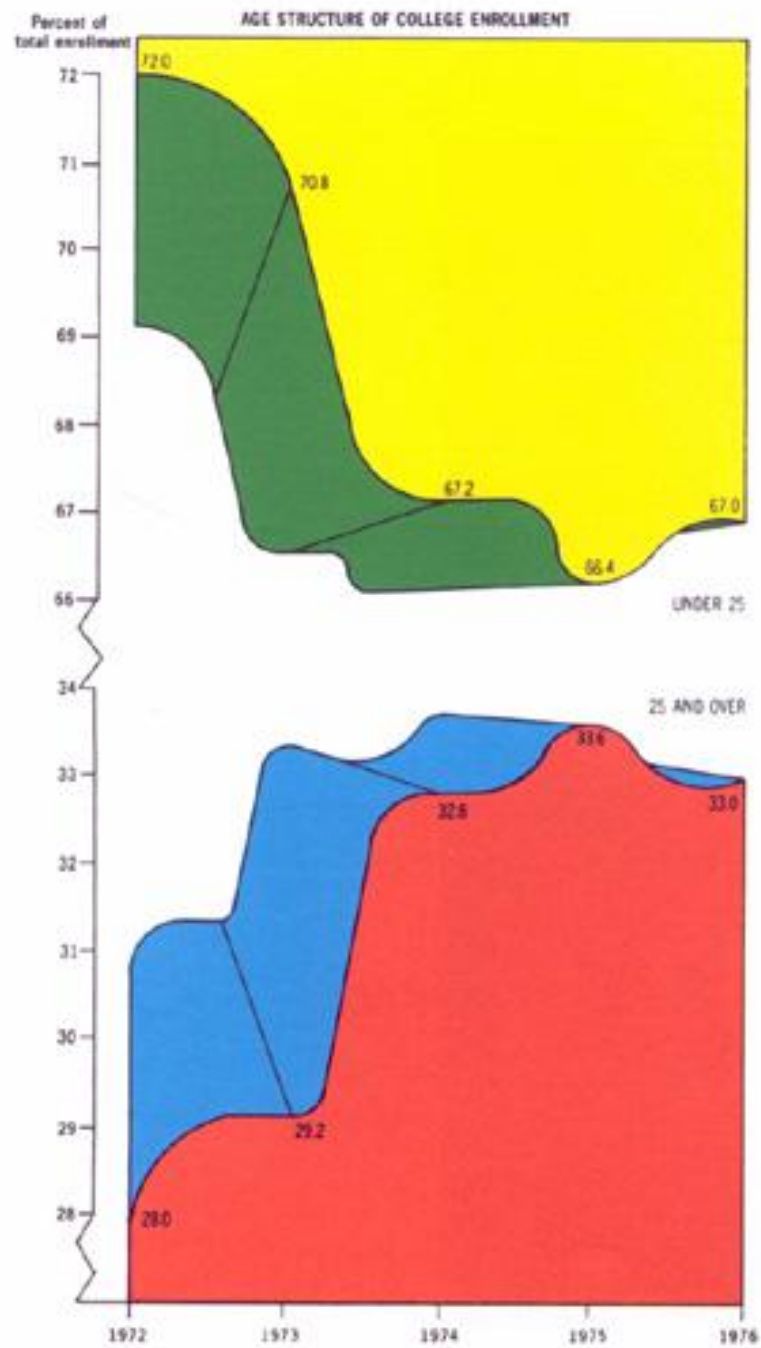
Women	Age	Men
Sweden	78	
	77	
France, US, Japan, Canada	76	
Finland, Austria, UK	75	
USSR, Germany	74	
	73	
	72	Sweden
	71	Japan
	70	
	69	Canada, UK, US, France
	68	Germany, Austria
	67	Finland
	66	
	65	
	64	
	63	USSR
	62	

Source: *Social Indicators III*

Table 2/29

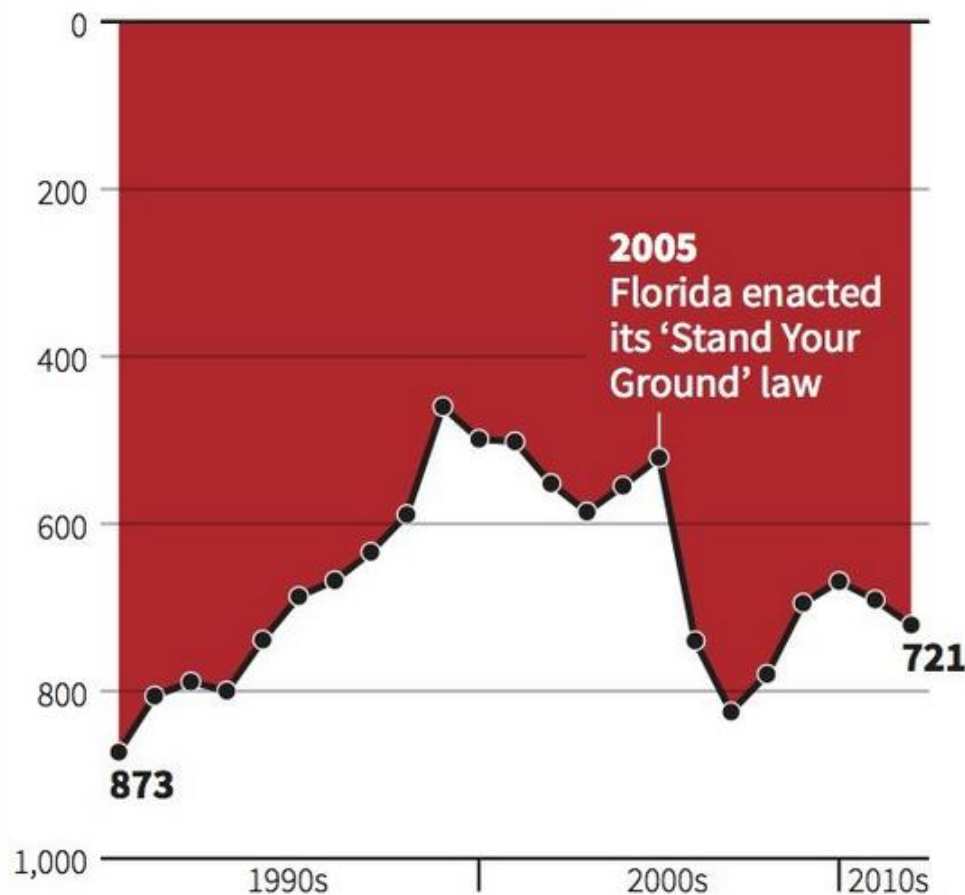
**THE WORST GRAPH EVER TO
FIND ITS WAY INTO PRINT.**

AMERICAN EDUCATION



Gun deaths in Florida

Number of murders committed using firearms



Source: Florida Department of Law Enforcement

C. Chan 16/02/2014

REUTERS

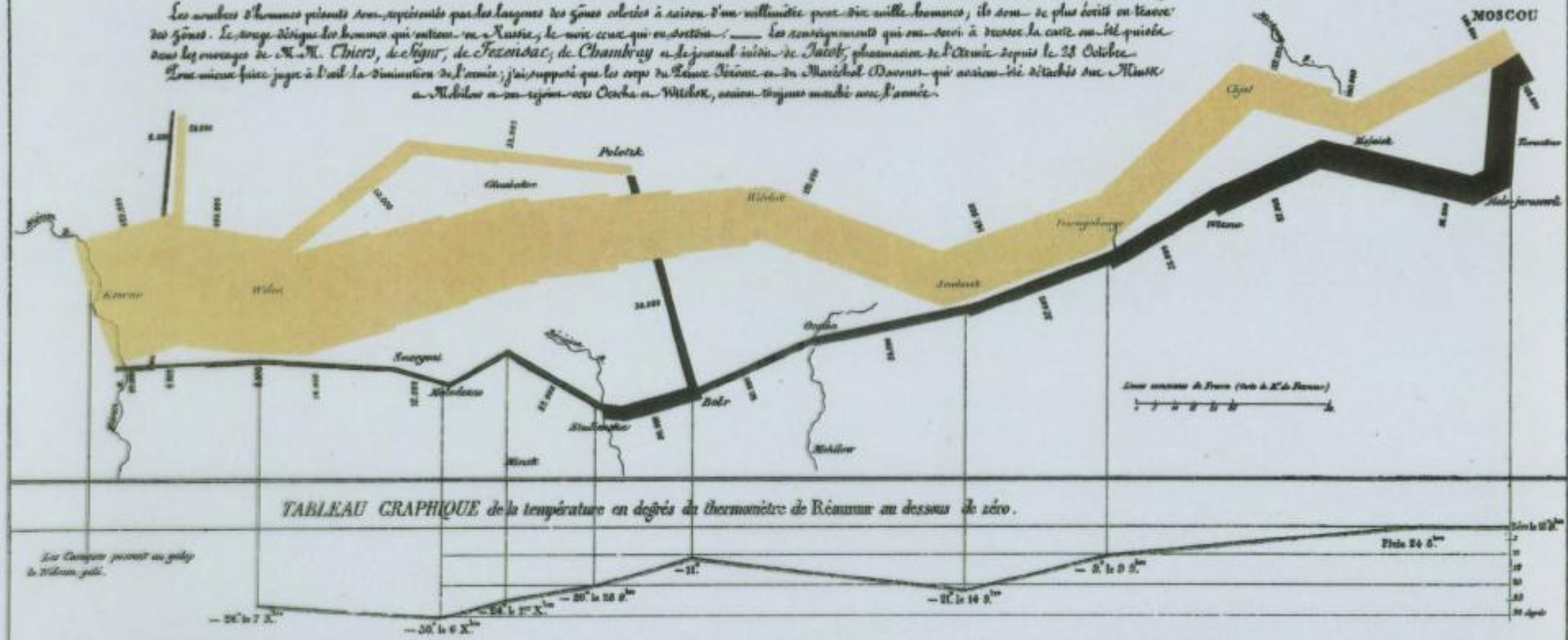
THE WORLD'S CHAMPION GRAPH

**DESIGNED AND DRAWN BY CHARLES
JOSEPH MINARD
NOVEMBER 20, 1869**

Carte Figurative des pertes successives en hommes de l'Armée Française dans la Campagne de Russie 1812-1813.

Dressée par M. MINARD, Inspecteur Général des Ponts et Chaussées en retraite. Paris, le 20 Novembre 1869

Les nombres d'hommes présents sont représentés par les largeurs des zones colorées à raison d'un millimètre pour dix mille hommes; ils sont de plus écrits en lettres des zones. Le rouge désigne les hommes qui entrent en Russie, le noir ceux qui en sortent. Les renseignements qui ont servi à dresser la carte ont été puisés dans les ouvrages de M. M. Chiers, de Segur, de Fozzard, de Chambray et le journal inédit de Jacot, pharmacien de l'Armée depuis le 28 Octobre. Tout mieux faire juger à l'œil la diminution de l'armée; j'ai supposé que les corps du Prince Jérôme et du Maréchal Davout qui avaient été détachés sur Minsk et Mielow n'en rejoignent pas Oczka et Wilna, mais qu'ils marchent avec l'armée.



CHAPTER 4, FIGURE 1. Charles Joseph Minard's famous plot of Napoleon's Russian Campaign. (Reprinted, with permission, from Tufte, 1983.)